Group Theory and the Quintic : Historical Notes

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1 Fundamental Theorem of Algebra

Definition A polynomial over a field F is a *rational* and *integral* expression of the form

$$\mathbf{I} \quad f(x) = a_n x^n + a_{n-1}x^{n-1} + ... + a_1 x + a_0, \tag{1}$$

where the coefficients $a_i \in F, i = 1, 2, \ldots, n$ and *rational* means free from fractional powers and *integral* means never in the denominator of a fraction.

Another commonly used definition of polynomials used the literature is:

$$\mathbf{II} \quad f(x) = a_0 x^n + a_1 x^{n-1} + ... + a_{n-1}x + a_n, \tag{2}$$

The difference involves the numbering scheme used on the coefficients:

$$a_j^{\mathbf{II}} = a_{n-j}^{\mathbf{I}}$$

A polynomial function is a function which can be written either in the form of Eq. (1) or Eq. (2).

Theorem 1.1 (Eisenstein's Irreducibility Criterion) *For a given prime p, let $a(x) = a_n x^n + a_{n-1}x^{n-1} + \ldots a_0$ be a polynomial with integral coefficients, such that $a_n \neq 0(\mod p)$, $\quad a_{n-1} = a_{n-2} = a_0 = 0(\mod p), a_0 \neq 0(\mod p^2)$. Then $a(x)$ is irreducible over the field of rational.*

Proof 1 Suppose

$$a(x) = gh,$$

where

$$\begin{aligned} g &= g_0 x^h + g_1 x^{h-1} + \ldots + g_{k-1}x + g_k \\ h &= h_0 x^l + h_1 x^{l-1} + \ldots + h_{l-1}x + h_l \end{aligned}$$

Then g and h have integral coefficients since f has. Comparing coefficients, we set

$$\begin{aligned} a_0 &= g_k h_l \\ a_1 &= g_k h_{l-1} + g_{k-1} h_l \\ a_2 &= g_k h_{l-2} + g_{k-1} h_{l-1} + \ldots + g_{k-2} h_l \end{aligned}$$

$$\begin{array}{rcl}
\dots & & \dots \\
a_l & = & g_k h_{l-k} + g_{k-1} h_{l-k+1} + \dots + g_1 h_{l-1} + h_l \\
\dots & & \dots,
\end{array}$$

where the h_i with possible negative subscripts are 0 and where

$$h_0 = 1.$$

if now every a_i, other than a_0 is divisible by p and a_n not by p^2, we conclude from the first equation that either g_k or h_l is divisible by p and the other is not. But when g_k is divisible by p, we conclude from the following equations that $g_{k-1}, g_{k-2}, \dots, g_1$ are divisible by p, and from the last equation that h_l is also divisible by p which is a contradiction from the assumption that a_n is not divisible by p^2.

Proof 2 In any possible factorization $(n = m + k)$

$$a(x) = bc = (b_n x^n + b_{n-1} x^{n-1} + \dots + b_1 x + b_0)(c_k x^k + c_{k-1} x^{k-1} + \dots + c_1 x + c_0)$$

we may assume that both factors have integral coefficients. Since $a_0 = b_0 c_0$ the hypothesis $a_0 \not\equiv 0 \pmod{p^2}$ means that not both b_0 and c_0 are divisible by p. Suppose $b_0 \not\equiv 0 \pmod p$, while $c_0 \equiv 0 \pmod p$. But $b_m c_k = a_n \not\equiv 0 \pmod p$, so $c_k \not\equiv 0 \pmod p$. Pick the smallest $r \leq k$ for which $c_r \not\equiv 0 \pmod p$, with $c_{r-1} \equiv \dots \equiv c_0 \equiv 0 \pmod p$. Then

$$a_r = b_0 c_r + b_1 c_{r-1} + \dots + b_r c_0 = b_0 c_r \pmod p.$$

But $\boldsymbol{b}_0 \not\equiv 0 \pmod p$ and $c_r \not\equiv 0 \pmod p$ give $a_r \not\equiv 0 \pmod p$, for p is a prime. By the hypothesis, the only coefficient a_r for which this is possible is a_n, therefore $r = n$: the degree of the second of the proposed factors must be n (the same as degree of $a(x)$, so that $a(x)$ is irreducible. ∎

This criterion can be applied to the cyclotomic polynomial

$$\phi(x) = \frac{x^p - 1}{x - 1} = x^{p-1} + x^{p-2} + \dots + x + 1. \tag{3}$$

This is the polynomial whose roots are the primitive p-th roots of unity. The Eisenstein criterion does not apply to Eq. (3) as it stands, but a simple change of variables $y = x - 1$ does the trick, for the binomial expansion gives

$$\frac{x^p - 1}{x - 1} = \frac{(y+1)^p - 1}{y} = y^{p-1} + p y^{p-2} + \frac{p(p-1)}{1 \cdot 2} y^{p-3} + \dots + p.$$

The binomial coefficients which appear on the right are all integers divisible by the prime p, for p occurs in each numerator as a factor and can never be cancelled out by the smaller integers in the denominator. The polynomial in y satisfies the hypothesis of the Eisenstein criterion and hence is irreducible. Therefore the original cyclotomic polynomial, Eq. (3), $\phi(x)$ is irreducible.

Theorem 1.2 (Extreme Value Theorem for real-valued functions of two real variables:) *Let D be the* **closed disk** *or radius $r > 0$ and centered at the origin, i.e,*

$$D = \{(x_1, x_2) \in \mathbb{R}^2 | x_1^2 + x_2^2 \leq r\}.$$

Let $f : D \to \mathbb{R}$ be a continuous function on the closed disk $D \subset \mathbb{R}^2$. Then f is bounded and attains its minimum and maximum values on D. In other words, there exists points $x_m, x_M \in D$ such that

$$f(x_m) \leq f(x) \leq f(x_M)$$

for every possible choice of point $x \in D$.

Theorem 1.3 (Euler-Gauss: Fundamental Theorem of Algebra) *Every polynomial $f(x) = a_n x^n + a_{n-1}x^{n-1} + \ldots + a_1 x + a_0$ of positive degree with complex coefficients has a complex root.*

We can regard $(x, y) \to |f(x + iy)|$ as a function $\mathbb{R}^2 \to \mathbb{R}$. We denote this function by $|f(\cdot)|$ or $|f|$. Since polynomials and the square root function are continuous, then $|f|$ is also continuous.

Lemma 1.4 *Let $f : \mathbb{C} \to \mathbb{C}$ be any polynomial function. Then there exists a point $z_0 \in \mathbb{C}$ where the function $|f|$ attains a minimum value in $\mathbb{R}$.*

Proof If f is a constant polynomial function, then the statement of Lemma (1.4) is trivial, so we may take $z_0 = 0$.
If f is not constant, then the degree of the polynomial defining f is at least one. In this case, we can denote f expliciitly as Eq. (1) with $a_n \neq 0$. Now assume that $z \neq 0$, and set $A = max\{|a_0|, |a_1|, \ldots, a_{n-1}|\}$. We can obtain a lower bound for $|f(z)$ as follows:

$$\begin{aligned} |f(z)| &= |a_n||z^n||1 + \frac{a_{n-1}}{a_n z} + \ldots + \frac{a_0}{a_n z^n}| \\ &\geq |a_n||z^n|| \left(1 - \frac{A}{|a_n|} \sum_{k=1}^{\infty} \frac{1}{|z|^k}\right) = |a_n||z^n|| \left(1 - \frac{A}{|a_n|} \frac{1}{|z| - 1}\right) \end{aligned}$$

For all $z \in \mathbb{C}$ such that $|z|| \geq 2$, we can further simplify this expression to

$$|f(z)| \geq |a_n||z^n| \left(1 - \frac{2A}{|a_n||z||}\right).$$

It follows from this inequality that there is an $r > 0$ such that $|f(z)| > |f(0)|$, for all $z \in \mathbb{C}$ satisfying $|z| > r$. Let $D \subset \mathbb{R}^2$ be the disk of radius r centered at 0, and define a function $g : D \to \mathbb{R}$, by

$$g(x, y) = |f(x + iy)|.$$

Since g is continuous, we can apply Thm (1.2) in order to obtain a point $(x_0, y_0) \in D$ such that g attains its minimum at (x_0, y_0). By the choice of r we have that for $z \in C\ D, |f(z)| > |g(0, 0)| \geq |g(x_0, y_0)|$. Therefore $|f|$ attains its minimum at $z = x_0 + iy_0$.

Proof of Thm (1.3): We rely on the fact that the function $|f(x)|$ attains its minimum value by Lemma (1.4). Let $z_0 \in \mathbb{C}$ be a point where the minimum is attained. We will show that if $f(z_0) \neq 0$, then z_0 is not a minimum value, thus proving that the minimum value of $|f(z)|$ is zero and therefore $f(z_0) = 0$.

If $f(z_0) \neq 0$, then we can define a new function $g : \mathbb{C} \to \mathbb{C}$ by setting

$$g(z) = \frac{f(z + z_0)}{f(z_0)}, \text{ for all } z \in \mathbb{C}.$$

Not that g is a polynomial of degree n, and that the minimum of $|f|$ is attained at z_0 if and only if the minimum of $|g|$ is attained at $z = 0$. Moreover, it is clear that $g(0) = 1$. Therefore g is a polynomial of the form

$$g(z) = b_n z^n + \ldots + b_k z^k + 1,$$

with $n \geq 1$ and $b_k \neq 0$ for some $1 \leq k \leq n$. Let $b_k = |b_k| e^{i\theta}$, and consider z of the form

$$z = r|b_k|^{-1/k}\, e^{i(\pi-\theta)/k}, \tag{4}$$

with $r > 0$. For z of this form we have

$$g(z) = 1 - r^k + r^{k+1} h(r),$$

with h a polynomial. Then for $r < 1$, we have by the triangle inequality that

$$|g(z)| \leq 1 - r^k + r^{k+1}|h(r)|.$$

For $r > 0$ sufficiently small we have $r|h(r)| < 1$, by the continuity of the function $rh(r)$ and the fact that it vanishes at $r = 0$. Hence we find that

$$|g(z)| \leq 1 - r^k(1 - r|h(r)|) < 1,$$

for some z having the form given in Eq. (4) with $r \in (0, r_0)$ and $r > 0$ sufficiently small. But then the minimum of $|g| : \mathbb{C} \to \mathbb{R}$ cannot possibly be equal to 1. ∎

2 Rational Roots

Theorem 2.1 *If the equation with integral coefficients*

$$a_n x^n + a_{n-1} x^{n-1} + \ldots + a_0 = 0 \tag{5}$$

has a rational root r/s, where r and s are integers without a common divisor other than unity, then r is a divisor of a_0 and s is a divisor of a_n.

Proof If x in Eq. (5) is replaced by r/s and the denominator is cleared of factors of s, then we obtain the following

$$a_n r^n + a_{n-1} r^{n-1} s + a_{n-2} r^{n-2} s^2 + \ldots + a_1 r s^{n-1} = -a_0 s^n.$$

And since r and s are relatively prime, it follows that r is a divisor or a_0, as was to be proved.

Corollary 2.2 *if the coefficients in the equation*

$$x^n + p_1 x^{n-1} + p_2 x^{n-2} + \ldots + p_n = 0$$

are integers, then every rational root of the equation is an integer and a divisor of the constant term p_n.

Proof This corollary follows since s can have only the values ± 1 if $a_n = 1$.

3 Cyclotomic Polynomials

Definition The cyclotomic polynomial $\phi(x)$ for non-trivial p-th roots of unity (other than $x = 1$) is given by

$$\phi(x) = \frac{(x^p - 1)}{x - 1} = x^{p-1} + x^{p-2} + \ldots x + 1.$$

Theorem 3.1 *$\phi(x)$ is an irreducible polynomial.*

Proof As it stands, the Eisenstein criterion for irreducibility doesn't apply. However a change of variable $y = x - 1$ and the binomial expansion changes the form into

$$\frac{(x^p - 1)}{x - 1} = \frac{((y+1)^p - 1)}{y} = y^{p-1} + py^{p-2} + \frac{p(p-1)}{1 \cdot 2} y^{p-3} + \ldots + p.$$

The binomial coefficients which appear on the right hand side are all integers divisible by the prime p, for p occurs in each numerator as a factor, and can never be cancelled out by the smaller integers in the denominator. The polynomial in y therefore satisfies the Eisenstein criterion and is therefore irreducible – which means that the original equation for $\phi(x)$ is irreducible. ∎

Theorem 3.2 (Irreducibility of $x^p - a = 0$) *A polynomial $f(x) = x^p - a = 0$ of prime degree p over a field K is either irreducible over K or has a root in K.*

Proof 1 *based on the Rational Root Theorem* Let α be a root of $f(x) = x^p - a$ (assume irreducible or not). Let $K = \mathbb{Q}(\alpha)$. (Assume we're working over $\mathbb{Q}$ or $\mathbb{Z}$ by Gauss's lemma for monic polynomials).

Since p prime, $[K : \mathbb{Q}] = 1$ or p.

If =1, $\alpha \in \mathbb{Q}$, so rational root.

Hence if no rational root, $[K : \mathbb{Q}] = p$, so minimum polynomial has degree p, so f (monic) is the minimum polynomial, hence irreducible.

Proof 2 Adjoin to K a primitive p-th root of unity ξ and the a root u of $x^p - a$. The resulting extension $K(\xi, u)$ contains the p roots $u, \xi u, \xi^2 u, \ldots, \xi^{p-1} u$ of the polynomial $x^p - a$, hence is the root field of this polynomial, which has the factorization

$$x^p - a = (x - u)(x - \xi u)(x - \xi^2 u) \ldots (x - \xi^{p-1} u).$$

Suppose that $x^p - a$ has a proper factor $g(x)$ over K of positive degree $m < p$. This factor $g(x)$ is then a product of m of the linear factors of $x^p - a$ over $K(\xi, u)$, so that the constant term b in $g(x)$ is a product of the roots $\xi^i u$. Therefore $b = \xi^k u^m$, for some integer k, and

$$b^p = (\xi^k u^m)^p = (\xi^p)^k (u^p)^m = (u^p)^m = a^m.$$

From this we can find in K a p-th root of a, for $m < p$ is relatively prime to p and there exists integers s and t with $sm + tp = 1$, so that

$$b^p = a^m = a^{1-tp} = a/a^{tp},$$

and $a = (b^s a^t)^p$. The assumed reducibility of $x^p - a$ over K therefore yields a root $(b^s a^t)$ of $x^p - a$ in K. ∎

Theorem 3.3 (Abel's Version of the Irreducibility of $x^p - F$) *If the equations*

$$\begin{aligned} f_1x^{p-1} + f_2x^{p-2} + \ldots + f_p &= 0 \\ x^p - F &= 0, \end{aligned} \tag{6}$$

where p is a prime number, are simultaneously satisfied, either $f_1, f_2, f_3, \ldots, f_p$ all vanish, or else one of the roots of $x^p - F = 0$ can be expressed rationally in terms of $f_1, f_2, \ldots, f_p$ and F.

Proof For, suppose the coefficients $f_1, f_2, \ldots, f_p$ don't vanish. The the two equations will have a greatest common divisor (GCD) of the form

$$g(x) = x^k + g_1x^{k-1} + g_2x^{k-2} + \ldots + g_k = 0,$$

the coefficients of which are rational functions of $F, f_1, f_2, \ldots, f_p$. Now If x_1 be any one of the roots common to equations (6), the other roots will be of the form

$$\omega^\alpha x_1, \omega^\beta x_1, \ldots, \text{ where } \omega^p = 1.$$

and hence

$$g_k = x_1^k\omega^{\alpha+\beta}\ldots = \omega^\pi x_1^k.$$

Since p is a prime number we can find two numbers m and n of which one is negative, such that $mp + nk = 1$. Also

$$g_k^n = \omega^{n\pi}x_1^{\pi k} = \omega^{n\pi}x_1^{1-nk},$$

and therefore

$$\omega^{n\pi}x_1 = g_k^nF^m,$$

therefore $\omega^{n\pi}x_1$, which is a root of Eq. (6) is expressed rationally in terms of $F, f_1, f_2, \ldots f_k$.

4 Determinants and Their Properties

Consider a set of vectors in an n-dimensional Vector Space: $V = \{\boldsymbol{a}_1, \boldsymbol{a}_2, ..., \boldsymbol{a}_n\}$ in $\mathbb{R}^n$.

Determinant The Determinant $D(\boldsymbol{a}_1, \boldsymbol{a}_2, ..., \boldsymbol{a}_n)$ has the following properties

i) Invariance property: $D(\boldsymbol{a}_1, \boldsymbol{a}_2, ..., \boldsymbol{a}_n)$ is to remain unchanged if some $\boldsymbol{a}_i$ is replaced by $\boldsymbol{a}_i + \boldsymbol{a}_j$ $(i \neq j)$;

ii) Homogeneity property: $D(\boldsymbol{a}_1, \boldsymbol{a}_2, ..., \boldsymbol{a}_n)$ is to go over into $\lambda D(\boldsymbol{a}_1, \boldsymbol{a}_2, ..., \boldsymbol{a}_n)$ if $\boldsymbol{a}_i$ is replaced by $\lambda\boldsymbol{a}_i$;

iii) the Normalization: $D(\boldsymbol{e}_1, \boldsymbol{e}_2, ..., \boldsymbol{e}_n) = 1$.

The determinant is the unique function of vectors $\{\boldsymbol{a}_1, \boldsymbol{a}_2, ..., \boldsymbol{a}_n\}$ that satisfy the above properties [3].

Expansion of Determinant Let the S_n be the set of all permutation of the numbers $\{1, 2, ..., n\}$. In the case of n-by-n matrix A we obtain the complete development of the determinant of A as follows:

$$D(\boldsymbol{a}_1, \boldsymbol{a}_2, ..., \boldsymbol{a}_n) = \det|A| = \sum_{\sigma \in S_n} \epsilon_{\sigma_1\sigma_2\ldots\sigma_n} a_{1\sigma_1}a_{2\sigma_2}...a_{n\sigma_n} = \begin{vmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \ldots & \ldots & \ldots & \ldots \\ a_{i1} & a_{i2} & \ldots & a_{in} \\ \ldots & \ldots & \ldots & \ldots \\ a_{n1} & a_{n2} & \ldots & a_{nn} \end{vmatrix} \tag{7}$$

If the rows are permuted by $\rho \in S_n$, we find

$$\epsilon_{\rho_1\rho_2\ldots\rho_n} \det|A| = \sum_{\sigma\in S_n} \epsilon_{\sigma_1\sigma_2\ldots\sigma_n} a_{\rho_1\sigma_1} a_{\rho_2\sigma_2} \ldots a_{\rho_n\sigma_n} \tag{8}$$

Cofactor of an element a_{rs},

$$M_{rs} = \text{cofactor}(a_{rs}) = \sum_{\sigma\in S_n,\ \sigma_r=s} \epsilon_{\sigma_1\sigma_2\ldots\sigma_n} a_{1\sigma_1} a_{2\sigma_2} \ldots a_{r-1\sigma_{r-1}} a_{r+1\sigma_{r+1}} \ldots . a_{n\sigma_n} \tag{9}$$

Relation between cofactors and minors obtained by taking the determinant obtained from A by striking out row r and column s from A. We denote such minor determinants by A_{rs} :

$$M_{rs} = (-1)^{r+s} A_{rs}$$

For every r, $1 \le r \le n$ we can write $\det|A|$ in terms of a cofactor expansion

$$\det|A| = \sum_{s=1}^{n} a_{rs}\, M_{rs} = \sum_{s=1}^{n} (-1)^{r+s}\, a_{rs}\, A_{rs} \tag{10}$$

Also, if there is a mismatch between the row indices between the matrix element and the cofactor or minor, i.e.,

$$\sum_{s=1}^{n} a_{ks}\, M_{rs} = \sum_{s=1}^{n} (-1)^{r+s}\, a_{ks}\, A_{rs} = 0, \text{ for } \quad k \ne r \tag{11}$$

The case $k \ne r$ corresponds to a determinant with two identical rows and hence vanishes because the rows are linearly dependent. Putting both of the above equations together we find

$$\sum_{s=1}^{n} a_{ks}\, M_{rs} = \sum_{s=1}^{n} (-1)^{r+s}\, a_{ks}\, A_{rs} = \delta_{kr} \det|A| \tag{12}$$

Classical Adjoint and Inverse Matrix We define the Classical Adjoint, also known as the Adjugate, of the matrix A as

$$\mathbf{adj}(A) = M^\dagger,$$

where $M^\dagger$ is the transpose of the cofactor matrix, M. Using Eq.(12), the inverse of A is given by

$$A^{-1} = \frac{1}{|A|}\mathbf{adj}.(A) = \frac{1}{|A|} M^\dagger.$$

Since $A^{-1}A = I$ we see that the determinant of A^{-1} is $|A|^{-1}$. Therefore the determinant of the Classical Adjoint matrix is

$$|\mathbf{adj}(A)| = |A|^n ||A|^{-1} = |A|^{n-1},$$

for an n-by-n matrix A.

Left Inverse and Right Inverse Suppose we have a left inverse X and a right inverse Y of a matrix A. This means that

$$XA = AY = I.$$

Therefore we have

$$\begin{aligned} XAY &= X, \quad \text{and} \\ XAY &= Y. \end{aligned}$$

Therefore

$$X = XAY = Y, \text{ or } X = Y.$$

Therefore left and right inverses of a non-singular matrix must be equal.

Laplace's General Expansion Theorem: Divide the set of integers $1, 2, ..., n$ into two complementary sets $\{\alpha_1, \alpha_2, ..., \alpha_p\}$ and $\{\beta_1, \beta_2, ..., \beta_q\}$ where $n = p + q$ and

$$\alpha_1 < \alpha_2 < ... < \alpha_p \text{ and } \beta_1 < \beta_2 < ... < \beta_q.$$

These two sets remained fixed throughout the following:

$$\det A = \sum_{\substack{\rho_1, \rho_2, ..., \rho_p \\ \rho_1 < \rho_2 < ... < \rho_p}} (-1)^{(\alpha_1 + \alpha_2 + ... + \alpha_p + \rho_1 + \rho_2 ... + \rho_p)} \begin{vmatrix} a_{\alpha_1\rho_1} & a_{\alpha_1\rho_2} & \cdots & a_{\alpha_1\rho_p} \\ a_{\alpha_2\rho_1} & a_{\alpha_2\rho_2} & \cdots & a_{\alpha_2\rho_p} \\ \cdots & \cdots & \cdots & \cdots \\ a_{\alpha_i\rho_1} & a_{\alpha_i\rho_2} & \cdots & a_{\alpha_i\rho_p} \\ \cdots & \cdots & \cdots & \cdots \\ a_{\alpha_p\rho_1} & a_{\alpha_p\rho_2} & \cdots & a_{\alpha_p\rho_p} \end{vmatrix} \begin{vmatrix} a_{\beta_1\sigma_1} & a_{\beta_1\sigma_2} & \cdots & a_{\beta_1\sigma_q} \\ a_{\beta_2\sigma_1} & a_{\beta_2\sigma_2} & \cdots & a_{\beta_2\sigma_q} \\ \cdots & \cdots & \cdots & \cdots \\ a_{\beta_i\sigma_1} & a_{\beta_i\sigma_2} & \cdots & a_{\beta_i\sigma_q} \\ \cdots & \cdots & \cdots & \cdots \\ a_{\beta_q\sigma_1} & a_{\beta_q\sigma_2} & \cdots & a_{\beta_q\sigma_q} \end{vmatrix} \tag{13}$$

The sets $\{\alpha_1, \alpha_2, ..., \alpha_p\}$ and $\{\beta_1, \beta_2, ..., \beta_q\}$ were defined to be fixed. The summation is taken over all p-tuples $(\rho_1, \rho_2, ..., \rho_p)$ from among the numbers $1, 2, ..., n$ for which $\rho_1 < \rho_2 < ... < \rho_p$ and where

$$\sigma_1 < \sigma_2 < ... < \sigma_q$$

are the remaining q integers.

5 Vandermonde Determinant

The Vandermonde Determinant for n variables $x = [x_1, x_2, \ldots, x_n]$ is given by

$$V_n(x) = \begin{bmatrix} 1 & x_1 & x_1^2 & x_1^3 & \ldots & x_1^n \\ 1 & x_2 & x_2^2 & x_2^3 & \ldots & x_2^n \\ 1 & x_3 & x_3^2 & x_3^3 & \ldots & x_2^n \\ \vdots & & & & & \\ 1 & x_n & x_n^2 & x_n^3 & \ldots & x_n^n \end{bmatrix}$$

Except for the first column, subtract x_1 times the previous column to the rest of the columns. Since a multiple of any column added to any other column does not change the determinant, we can write

$$\det V_n(x) = \det \begin{bmatrix} 1 & 0 & 0 & 0 & \ldots & 0 \\ 1 & x_2 - x_1 & x_2(x_2 - x_1) & x_2^2(x_2 - x_1) & \ldots & x_2^{n-1}(x_2 - x_1) \\ 1 & x_3 - x_1 & x_3(x_3 - x_1) & x_3^2(x_3 - x_1) & \ldots & x_3^{n-1}(x_3 - x_1) \\ \vdots & & & & & \\ 1 & x_n - x_1 & x_n(x_n - x_1) & x_n^2(x_n - x_1) & \ldots & x_n^{n-1}(x_n - x_1) \end{bmatrix}$$

Expanding the determinant using the elements in the first row we have

$$\det V_n(x) = \det \begin{bmatrix} x_2 - x_1 & x_2(x_2 - x_1) & x_2^2(x_2 - x_1) & \dots & x_2^{n-1}(x_2 - x_1) \\ x_3 - x_1 & x_3(x_3 - x_1) & x_3^2(x_3 - x_1) & \dots & x_3^{n-1}(x_3 - x_1) \\ \vdots & & & & \\ x_n - x_1 & x_n(x_n - x_1) & x_n^2(x_n - x_1) & \dots & x_n^{n-1}(x_n - x_1) \end{bmatrix}$$

Factoring out $x_2 - x_1$ from the first row, $x_3 - x_1$ from the second row, and so on until the last row we have

$$\det V_n(x_1, x_2, \dots, x_n) = [\prod_{i=2}^{n}(x_i - x_1)] \det \begin{bmatrix} 1 & x_2 & x_2^2 & \dots & x_2^{n-1} \\ 1 & x_3 & x_3^2 & \dots & x_3^{n-1} \\ \vdots & & & & \\ 1 & x_n & x_n^2 & \dots & x_n^{n-1} \end{bmatrix} = \prod_{i=2}^{n}(x_i - x_1) \ \det V_{n-1}(x_2, x_3, \dots, x_n)$$

Proceeding in a similar way by subtracting x_2 times the previous column except for the first column we can continue

$$\det V_n(x_1, x_2, \dots, x_n) = [\prod_{i=2}^{n}(x_i - x_1)]\, [\prod_{i=3}^{n}(x_i - x_2)] \ \det V_{n-2}(x_3, x_4, \dots, x_n)$$

To arrive at

$$\det V_n(x_1, x_2, \dots, x_n) = \prod_{\substack{1<j \\ i<j}}^{n}(x_j - x_i) = \prod_{1 \le i < j \le n}(x_j - x_i)$$

6 Roots of Unity

Let r denote an nth root of unity. If there is no positive integer $m < n$ such that $r^m = 1$, then we say that r is a primitive root nth root of unity.

Theorem 6.1 *Let $R = \cos(2\pi/n) + i\sin(2\pi/n)$. The number R^k, $(1 \le k \le n)$ is a primitive nth roof of unity, if and only if, the integers k and n are relatively prime.*

Proof First assume that k and n have a common factor d. Then $n = sd, k = td$. Then

$$(R^k)^s = (R^{td})^s = (R^{sd})^t = (R^n)^t = 1.$$

Since $s < n$, and $(R^k)^s = 1$, it follows that R^k is not a primitive root of unity.

We next assume that k and n are relatively prime, and show that $(R^k)^n$ cannot have the value 1 if $m < n$. We use De Moivre's theorem to write

$$(R^k)^m = \cos\frac{2km\pi}{n} + i\sin\frac{2km\pi}{n}.$$

If the latter quantity is equal to 1, then $2km\pi/n$ must be a multiple of 2π, say $(2km\pi/n) = 2h\pi$. Therefore km must be divisible by n. But this is impossible since k has no factor in common with n and $m < n$ by assumption. Therefore $(R^k)^m \neq 1$ if $m < n$. Therefore R^k is a primitive root of unity if k and n are relatively prime.

7 Algebraic Form for a Rational Expression of a Radical

Let R be a rational number which is not a perfect p-th root and let p be a prime number so we don't have to further reduce the expression (composite numbers can always be factored into primes which can subsequently be considered separately by nesting the radical expressions). Let $\alpha = \sqrt[p]{R}$ and consider the set $\{\alpha, \alpha^2, \ldots, \alpha^{(p-1)}\}$. These components form the basis set for a vector space, V over a field F. The general form for an element in this space is given by

$$f(\alpha) = c_0 + c_1\alpha + \cdots + c_{p-1}\alpha^{p-1}, \quad c_i \in F, i = 0, 1, 2 \ldots, p-1. \tag{14}$$

A rational expression in V takes the form $P(\alpha)/Q(\alpha)$, where

$$\begin{aligned} P(\alpha) &= a_0 + a_1\alpha + \cdots + a_m\alpha^m \\ Q(\alpha) &= b_0 + b_1\alpha + \cdots + b_n\alpha^n, \end{aligned}$$

where $a_i \in F$ for $i = 0, 1, 2, \ldots,, m$ and $b_j = 0, 1, 2, \ldots n$.

Such a rational expression can be brought back into the form of $c_0 + c_1\alpha + \cdots + c_{p-1}\alpha^{p-1}$, by first using Eq. (14) for any terms with degree greater than p in either the numerator or denominator and by rationalizing the denominator by multiplying $Q(\alpha)$ by all of the conjugate values using the p-roots of unity ω: $\omega^p = 1$.
We put this in the form of:

Theorem 7.1 *Any algebraic function can be put in the form*

$$y = c_0 + c_1\alpha + c_2\alpha^2 + \ldots + c_{p-1}\alpha^{p-1},$$

for p a prime and $c_0, c_1, \ldots c_{p-1}, R$ are algebraic functions of lower order than y.

Proof We start by showing that it is possible to clear the denominator of terms that depend only on rational quantities and a free of roots of unity. To proceed we multiply the numerator and denominator by the conjugate values of $Q(\alpha)$ over the p roots of unity:

$$\frac{P(\alpha)}{Q(\alpha)} = \frac{P(\alpha)Q(\omega\alpha)Q(\omega^2\alpha)\ldots Q(\omega^{p-1}\alpha)}{Q(\alpha)Q(\omega\alpha)Q(\omega^2\alpha)\ldots Q(\omega^{p-1}\alpha)}.$$

In order to see how this works we examine the denominator on the right-hand side to see if any irrationalities involving α or roots of unity ω are left in the denominator:

$$D = Q(\alpha)Q(\omega\alpha)Q(\omega^2\alpha)\ldots Q(\omega^{p-1}\alpha).$$

Some of the books and papers suggest proceeding by using the distributive law to expand the product of polynomial functions on the right hand side and using the appropriate cyclotomic polynomial:

$$1 + \omega + \omega^2 + \ldots + \omega^{(p-1)} = 0,$$

to eliminate terms involving irrational terms in α or roots of unity. This brute force method appears very laborious to perform and we look for other methods. It is certainly true the the final expression for R is symmetric in the roots of the polynomial $Q(x)$ and hence by the theorem on Symmetric

Functions can be expressed in terms of the coefficients of $Q(x)$ and hence is rational. However, we would like probe in more detail to help visualize the problem.

Three such methods suggest themselves:

1) Use Viète's Formula for expanding a polynomial into linear factors and then use the properties of the roots of unity and separately calculate the products of the linear factors over the roots of unity.

2) Use Sylvester's Resolvent to eliminate x between $x^p - d = 0$ and $Q(x)$.

3) Use Twisted Circulant Matrices and calculate the determinant using the product of all the eigenvalues $\lambda_i = Q(\omega^i \alpha), i = 0, 1, 2, \ldots, (p-1)$,

Method A – Based on products over all the Roots of Unity

Consider the monic polynomial $Q(x) = x^p + a_{p-1}x^{p-1} + \ldots + a_1 x + a_0$, with roots $\alpha_0, \alpha_2, \ldots, \alpha_{p-1}$. By Viète's theorem,

$$\begin{aligned} Q(x) &= (x-\alpha_0)(x-\alpha_2)(x-\alpha_3)\ldots(x-\alpha_{p-1}) \\ &= \prod_{j=0}^{p-1}(x-\alpha_j) \end{aligned}$$

From

$$x^p - 1 = \prod_{i=0}^{p-1}(x-\omega^i)$$

we can show

$$x^p - d = \prod_{i=0}^{p-1}(x-\omega^i \sqrt[p]{R})$$

Taking the product of $f(x)$ over all the roots of $x^p - d = 0$ we form

$$\prod_{i=0}^{p-1} Q(\omega^i \sqrt[p]{R}) = \prod_{i=0}^{p-1} \prod_{j=0}^{p-1} (\omega^i \sqrt[p]{R} - \alpha_j) = \prod_{j=0}^{p-1} \prod_{i=0}^{p-1} (\omega^i \sqrt[p]{R} - \alpha_j) = \prod_{j=0}^{p-1} (d - \alpha_j^p)$$

The final expression on the right-hand side depends only on rational quantities and is free of the roots of unity.

Method B – Based on Sylvester's Resolvent

Let

$$\begin{aligned} f(x) &= x^p - d \\ Q(x) &= b_n \alpha^n + b_{n-1}\alpha^{n-1} + \cdots + b_1 \alpha + b_0, \end{aligned}$$

where R is not a p-th power of an element in the field of coefficients. $f(x)$ and $Q(x)$ will have a common root if the Sylvester Resultant formed by these two functions vanishes. The roots of $f(x)=0$ take the form $z_k = \omega^k \sqrt[p]{R}$ and we're assuming that $d^{\frac{1}{p}}$ is an irrationality in the domain containing the coefficients. Also with respect to Abel's rationalization method we need to show that

$$\prod_{k=0}^{p-1} Q(z_k) = G(z^p) = G(d) \text{ where } G(d) \text{ is rational in the domain of the coefficients of } Q(x). \tag{15}$$

Using $f(x)$ and $Q(x)$ we construct the Sylvester Resultant

$$\det(M) = \prod_{k=0}^{p-1} Q(z_k) = \det \begin{bmatrix} 1 & 0 & \cdots & -R & & \\ 0 & 1 & \ddots & \cdots & \ddots & \\ \vdots & & \ddots & \cdots & \ddots & \\ 0 & \cdots & & 1 & \cdots & -R \\ b_n & & \cdots & b_0 & & \\ & & \ddots & \cdots & \ddots & \\ & & & b_n & \cdots & b_0 \end{bmatrix}.$$

Since the roots of $f(x) = x^p - R$ are exactly $z_k = \omega^k \sqrt[p]{R}$, substituting them into the formula yields

$$\det(M) = \prod_{k=0}^{p-1} Q(z_k).$$

Because computing the determinant $\det(M)$ only involves additions, subtractions, and multiplication of the matrix entries, the final polynomial expression outputted by the determinant can only contain powers of combinations of the terms in the Sylvester matrix. Therefore the resulting polynomial must be a function of $x^p = R$ and the coefficients and must therefore be a rational expression. Sylvester's Resolvent depends only on rational quantities and is also free of the roots of unity.

Method C – Based on Twisted Circulant Matrix

A twisted Circulant matrix (or a Twistulant matrix) is a square $p \times p$ matrix where each row is shifted cyclically to the right by one position, much like a traditional circulant matrix. However, when the final element wraps around to the beginning, it is multiplied by a specific complex scalar R.

1. The General $n \times n$ Twisted Circulant Matrix Definition Let $\alpha = \sqrt[n]{R}$ be an algebraic radical of degree n. The algebraic denominator expression to be cleared is defined as: $Q(\alpha) = \sum_{j=0}^{n-1} c_j \alpha^j = c_0 + c_1\alpha + c_2\alpha^2 + \cdots + c_{n-1}\alpha^{n-1}$ The associated $n \times n$ twisted circulant matrix C is structured as follows:

$$C = \begin{pmatrix} c_0 & c_1 & c_2 & \cdots & c_{n-2} & c_{n-1} \\ d \cdot c_{n-1} & c_0 & c_1 & \cdots & c_{n-3} & c_{n-2} \\ d \cdot c_{n-2} & d \cdot c_{n-1} & c_0 & \cdots & c_{n-4} & c_{n-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ d \cdot c_2 & d \cdot c_3 & d \cdot c_4 & \cdots & c_0 & c_1 \\ d \cdot c_1 & d \cdot c_2 & d \cdot c_3 & \cdots & d \cdot c_{n-1} & c_0 \end{pmatrix}$$

Eigenvectors: Let $\omega = e^{\frac{2\pi i}{p}}$ be the primitive p-th root of unity. The eigenvectors are independent of the specific matrix elements $c_0, c_1, \ldots, c_{p-1}$ and rely solely on p and α. They can be explicitly defined by a scaled Discrete Fourier Transform (DFT) basis. For $k = 0, 1, \ldots, p-1$, the k-th right eigenvector v_k of C is given by:

$$v_k = \begin{bmatrix} 1 \\ \omega^k \alpha \\ \omega^{2k} \alpha^2 \\ \vdots \\ \omega^{(n-1)k} \alpha^{(p-1)} \end{bmatrix}.$$

Eigenvalues: While a standard circulant matrix uses pure Fourier modes as eigenvectors, a R-circulant matrix's eigenvectors account for the scalar shift R. Each eigenvalue λ_k corresponds to the following exact formula:

$$\lambda_k = \sum_{j=0}^{p-1} c_j \left(\omega^k \alpha\right)^j$$

Uncoupled Diagonalization: Because all twisted circulant matrices commute with the twisted shift operator, they are simultaneously diagonalizable.

Standard Circulant: When $d = 1$, the matrix collapses into a standard circulant matrix, and the eigenvectors become the standard columns of the inverse DFT matrix, defined entirely by the roots of unity.

2. The Determinant (Field Norm Formula)

The determinant factors into the product of the field conjugates:

$$\det(C) = \prod_{k=0}^{p-1} \lambda_k = \prod_{k=0}^{p-1} \left(\sum_{j=0}^{n-1} c_j \left(\omega^k \alpha\right)^j \right)$$

Since the left-hand side involves the determinant of an $n \times n$ matrix whose elements are free of ω and $\alpha = \sqrt[p]{R}$, therefore expanding the product on the right-hand side eliminates all instances of ω and α. This determinant is guaranteed to be a purely rational number if all a_j and R are rational. It yields a purely rational polynomial given by the determinant of C:

$$\det(C) = \mathrm{Norm}_{\mathbb{Q}(\sqrt[p]{R})/\mathbb{Q}}(Q(\alpha)) \in \mathbb{Q}$$

3. Clearing the Denominator via Classical Adjoint of the Twisted Circulant Matrix To rationalize the fraction $\frac{1}{Q(\alpha)}$, use the matrix inversion property involving the Classical Adjoint matrix $\mathrm{adj}(C)$:

$$C \cdot \mathrm{adj}(C) = \det(C) \cdot I_n$$

The vector of algebraic basis elements is defined as:

$$\vec{\alpha} = \begin{pmatrix} 1 \\ \alpha \\ \alpha^2 \\ \vdots \\ \alpha^{p-1} \end{pmatrix}$$

By the construction of the twisted circulant matrix, we have the identity:

$$C\vec{\alpha} = Q(\alpha)\vec{\alpha}$$

Multiplying both sides by the Classical Adjoint matrix $\mathrm{adj}(C)$ yields:

$$\det(C)\vec{\alpha} = Q(\alpha) \cdot \mathrm{adj}(C)\vec{\alpha}$$

Extracting the first component (the top row) gives the exact rationalization identity:

$$\frac{1}{c_0 + c_1\alpha + \cdots + c_{p-1}\alpha^{p-1}} = \frac{\sum_{j=0}^{p-1} M_{0,j}\alpha^j}{\det(C)}$$

Where $M_{0,j}$ represents the signed $(0, j)$ cofactor of the first row of matrix C:

$$M_{0,j} = (-1)^j \det(C_{\mathrm{minor}(0,j)})$$

The product of the eigenvalues of the Twisted Circulant is equal to the determinant of the same matrix whose elements depend only on rational quantities and are also free of the roots of unity.

Conclusion

The Twisted Circulant C compresses the elimination task when compared to Sylvester's Resolvent. By embedding the reduction rule $x^p = d$ directly into the matrix shift operations, it avoids tracking the explicit polynomial degrees of $A(x)$, wrapping them into a much smaller, highly symmetric $p \times p$ matrix:

$$C = \begin{pmatrix} a_0 & a_1 & \dots & a_{p-1} \\ d \cdot a_{p-1} & a_0 & \dots & a_{p-2} \\ \vdots & \vdots & \ddots & \vdots \\ d \cdot a_1 & d \cdot a_2 & \dots & a_0 \end{pmatrix}$$

The Identity Mathematically, the relationship between their determinants is absolute:

$$\det(C) = \mathrm{Res}(A, B) = \det(S(A, B))$$

All three methods provide different approaches to rationalizing the denominator of a rational algebraic expression. The Twisted Circulant approach is more compact than Sylvester's Resolvent:

Table 1: Summary of Differences

Feature	**Twisted Circulant Matrix**	**Sylvester Resultant Matrix**
Matrix Size	Compact: $p \times p$	Large: $2p - 1 \times 2p - 1$
Mathematica Basis	Linear operator on the field basis $\{1, \alpha, \dots, \alpha^{p-1}\}$	Elimination tool for two distinct polynomial rings
Intrinsic Rule	Radical Reduction $x^p = R$ built in to the "twist"	Radical Reduction explicitly handled by inserting $-R$ into the rows
Bezout Identity	Obtained by using the Classical Adjoint $\mathrm{adj}(C)$	Obtained by solving the system $S^\dagger\vec{x} = \vec{e}$

8 Lagrange

Theorem 8.1 (Lagrange) *The order of a subgroup is a divisor of the order of the group.*

Proof Consider a group G of N and a subgroup H composed of the substitutions

$$h_1 = I, h2, h3, \ldots, h_P. \tag{16}$$

If G constains no further substitutions, then $N = P$ and the theorem is true. Now let G contain a substitution g_2 not in H. Then G also must contain

$$g_2, h_2 g_2, h_3 g_2, \ldots, h_P g_2. \tag{17}$$

This set are all distinct and all different from the substitutions (16), since $h_\alpha g_2 = h_\beta$ requires that $g_2 = h_\alpha^{-1} h_\beta =$ a substition of H contrary to the assumption that g_2 is distinct from the elements of H. Hence the substitutions combined now give $2P$ distinct substitutions of G. If there are no other substitutions in G, then $N = 2P$ and the theorem is true. If there are any other distrinct substitutions g_3 in G not in (16) or (17), then G contains

$$g_3, h_2 g_3, h_3 g_3, \ldots, h_P g_3. \tag{18}$$

Moreover, these substitutions are all different from (16) and (17), since $h_\alpha g_3 = h_\beta g_2$ requires that $g_3 = h_\alpha^{-1} h_\beta g_2$ and shall belong to (17), contrary to assumption. We now have $3P$ distinct substitutions of G. Either $N = 3P$ or else G contains a substitution g_4 not in the sets (16), (18), (18). In the latter case G contains the products

$$g_4, h_2 g_4, h_2 g_4, \ldots, h_P g_4, \tag{19}$$

all of which are distrinct and all different from the sets (16), (18), (18), so that we have $4P$ distinct substitutions. Proceeding in this way, we finally reach a last set of P substitutions

$$g_n u, h_2 g_n u, h_3 g_n u, \ldots, h_P g_\nu, \tag{20}$$

since the order of H is finite. Hence $N = \nu P$.

Definition The number ν is called the *index* of the subgroup H under G and is denoted by $\nu = [G : H]$.

Corollary 8.2 *The order of any group H of substitutions on n letters is a divisor of $n!$. H is a subgroup of the symmetric group G_n on n letters.*

Theorem 8.3 *The period of any substitution contained in a group G of order N is a divisor of N.*

Corollary 8.4 (Applied to the Symmetric Group) *The order of any substitution group on a set of n elements is an exact divisor of $N = n!$, the quotient being the number of distinct value of the corresponding multiple-valued function.*

Let us arrange the N substitutions of G in a rectangular array using the substitutions of a subgroup H in the first row:

$$
\begin{array}{lllll}
h_1 = I & h_2 & h_3 & \dots & h_P \\
g_2 & h_2g_2 & h_3g_2 & \dots & h_Pg_2 \\
g_3 & h_2g_3 & h_3g_3 & \dots & h_Pg_3 \\
\vdots & \dots & \dots & \dots & \vdots \\
g_\nu & h_2g_\nu & h_3g_\nu & \dots & h_Pg_\nu
\end{array} \tag{21}
$$

The $g_1 = I, g_2, g_3, \dots, g_\nu$ multiply the elements of H on the right hand side forming right cosets. Similarly, a rectangular array for the substitiions could be formed by using left costs.

Theorem 8.5 *If ψ is a rational funciton of $x_1, x_2, \dots, x_n$ belonging to a subgroup H of index ν, then ψ is ν-valued under the full set of substitutions of G.*

Proof Apply to ψ all the N substitutions of G arranged in a rectangular array as in (21). All the substitutions belonging to the same row give the same answer since

$$\psi_{h_i g_\alpha} = (\psi_{h_i})_{g_\alpha} = (\psi)_{g_\alpha} = \psi_{g_\alpha}.$$

Hence there result at most ν values. But, if

$$\psi_{g_\alpha} = \psi_{g_\beta}$$

then $\psi_{g_\alpha g_b eta^{-1}} = \psi$, so that $g_\alpha g_\beta^{-1} h_i \in H$, leaving ψ unaltered. Hence $g_\alpha = h_i g_\alpha$, contrary to the assumption made in forming the rectangular array.

Regarding the orders of a group H and a larger group G containing H as a subgroup; that is to say, the order r of H is an exact divisor of the order G which contains H as a subgroup, the quotient m being the number of distinct values of a multiple-valued function unaltered by the substitutions of G which are obtained by the substutions of H.

Definition The ν distinct functions $\psi, \psi_{g_2}, \psi_{g_3}, \dots, \psi_{g_\nu}$ are called **conjugate values** of ψ under G.

Theorem 8.6 *The ρ distinct values which a rational function $\psi(x_1, \dots, x_n)$ takes when operated on by all $n!$ substitutions are the roots of an equation of degree ρ whose coefficients are rational functions of the elementary symmetric functions.*

Formation of Functions of a Given Group

We define the Galois Function having $n!$ values for all the substiutionts of the symmetric group

$$\psi_1 = \alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3 + \dots \alpha_n x_n,$$

in which $\alpha_1, \alpha_2, \dots, \alpha_n$ are n distinct arbitrary constants. This function is called the Galois Function. If we use all the P substitutions of H to form the functions $\psi_1, \psi_2, \dots, \psi_P$ by the substitutions of H anysymmetric funciton of these values will be unaltered by the substitutions of H. In particular $\phi_1 = (y - \psi_1)(y - \psi_2) \dots (y - \psi_P)$ will be unaltered by the substutions of H, and altered to a different value by the substitutions not in H. The function ϕ_1 is not therefore unaltered by the substitutions of any wider group that contains H as a subgroup. Noting the forms of the expression of the sums of powers of the roots in terms of the coefficients, we see that instead of the coefficients of the powers of y in ϕ_1 we may take sums of powers of $\psi_1, \psi_2, \dots, \psi_P$ up to the P^{th} and deduce that one at least of such P sums of powers will provide a function unlatered by the substitutions of H and altered by any other substitution of the symmetric group.

Theorem 8.7 *Every integral symmetric function of the distinct values of any integral multiple-valued function of n elements is a symmetric function of the elements themselves.*

Proof Let $F(\phi_1, \phi_2, \ldots, \phi_\rho)$ be any rational integral symmetric function of the ρ-values. Any substitution S whatever affecting the elements applied to these ρ-values either leaves any function unchanged or replaces it by one of the others. No two of the resulting values can be equal, for if $\phi_i S$ were equal to $\phi_j S$ it would follow, by applying the substitution S^{-1}, that $\phi_i = \phi_j$ which is contrary to the assumptions. Consequently the ρ values of ϕ are reproduced by S in some order or other. The symmetric function F therefore remains unchanged by any substitutiont, and is consequently a symmetric element of the elements themselves.

Corollary 8.8 *The ρ distinct values of an integral multiple-valued function are roots of an equation whose coefficients are integral symmetric functions of the elements themselves.*

Theorem 8.9 (Lagrange Theorem on functions belonging to the same group) *Two rational functions belonging to the same group are rationally expressed each in terms of the other.*

Proof Let ϕ_1 and ψ_1 be two functions belonging to the same group

$$H = \{I, S_2, S_3, \ldots, S_r\},$$

of order r and degree n, each of these functions having ρ distinct values, where $r * \rho = N$. Any substitution not contained in H will convert ϕ_1 into another of its values, say ϕ_k and at the same time convert ψ_1 into ψ_k. By operating all possible substitutions, ρ pairs of values $\phi_1, \psi_1 : \phi_2, \psi_2 : \ldots \phi_\rho, \psi_\rho$ are obtained. Now, in the first place, the rational function

$$\sum \phi^i \psi^j = \phi_1^i \psi_1^j + \phi_2^i \psi_2^j + \ldots + \phi_k^i \psi_k^j + \ldots + \phi_\rho^i \psi_\rho^j \tag{22}$$

is clearly a symmetric function of the elements, for it appears by the same reasoning as before that any substitution acting on this function will merely permute the terms leaving the sum unaltered. Therefore $\sum \phi^i \psi^j$ is a symmetric function of the elements.

If we now take $j = 1$ and assign to i all the values $0, 1, 2, \ldots, \rho - 1$ in succession, we obtain the following ρ equations linear in $\psi_1, \psi_2, \ldots, \psi_\rho$:

$$\begin{array}{rcrcccrr}
\psi_1 & + & \psi_2 & + & \ldots & + & \psi_\rho = & T_0 \\
\phi_1\psi_1 & + & \phi_2\psi_2 & + & \ldots & + & \phi_\rho\psi_\rho = & T_1 \\
\phi_1^2\psi_1 & + & \phi_2^2\psi_2 & + & \ldots & + & \phi_\rho^2\psi_\rho = & T_2 \\
\ldots & \ldots & \ldots & \ldots & \ldots & \ldots & \ldots = & \ldots \\
\phi_1^{\rho-1}\psi_1 & + & \phi_2^{\rho-1}\psi_2 & + & \ldots & + & \phi_\rho^{\rho-1}\psi_\rho = & T_{\rho-1}
\end{array} \tag{23}$$

where $T_0, T_1, \ldots, T_{\rho-1}$ are all symmetric functions of $x_1, x_2, \ldots, x_n$.

We can write the above as a linear system

$$\begin{bmatrix} 1 & 1 & 1 & 1 & \ldots & 1 \\ \phi_1 & \phi_2 & \phi_3 & \phi_4 & \ldots & \phi_\rho \\ \phi_1^2 & \phi_2^2 & \phi_3^2 & \phi_4^2 & \ldots & \phi_\rho^2 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \phi_1^{\rho-1} & \phi_2^{\rho-1} & \phi_3^{\rho-1} & \phi_4^{\rho-1} & \ldots & \phi_\rho^{\rho-1} \end{bmatrix} \begin{bmatrix} \psi_1 \\ \psi_2 \\ \vdots \\ \vdots \\ \psi_\rho \end{bmatrix} = \begin{bmatrix} T_0 \\ T_1 \\ \vdots \\ \vdots \\ T_{\rho-1} \end{bmatrix} \tag{24}$$

Linear systems such as Eq.(24) can be solved using Cramer's rule. Alternatively, in order to understand better the relatonship between ϕ_i and ψ_i let us consider a method based on transforming non-homogenous linear systems into sets of corresponding homogeneous linear systems in each of the unknown parameters. To this effect consider the linear system [4].

$$\begin{array}{ccccccc} x & + & y & + & z & = & T_0 \\ \alpha x & + & \beta y & + & \gamma z & = & T_1 \\ \alpha^2 x & + & \beta^2 y & + & \gamma^2 z & = & T_2 \end{array} \tag{25}$$

By adding a fourth equation to the above we can isolate one of the variables, say y, and determine the functional dependence of y on β and the symmetric functions of the coefficients. Adding $y = y$ to the above system and converting it into a system of homogenous equations by introducing a 4th unknown w we obtain the system

$$\begin{array}{ccccccc} 0 & + & y & + & 0 & = & y \\ x & + & y & + & z & = & T_0 \\ \alpha x & + & \beta y & + & \gamma z & = & T_1 \\ \alpha^2 x & + & \beta^2 y & + & \gamma^2 z & = & T_2 \end{array} \tag{26}$$

The above system can be converted into a Homogeneous System by moving the final column to the left-hand side of the equal sign.

$$\begin{array}{ccccccccc} 0 & + & y & + & 0 & - & y & = & 0 \\ x & + & y & + & z & - & T_0 & = & 0 \\ \alpha x & + & \beta y & + & \gamma z & - & T_1 & = & 0 \\ \alpha^2 x & + & \beta^2 y & + & \gamma^2 z & - & T_2 & = & 0 \end{array} \tag{27}$$

Writing the above system using a matrix of coefficients and introducing the 4th undetermined variable we find

$$\begin{bmatrix} 0 & 1 & 0 & -y \\ 1 & 1 & 1 & -T_0 \\ \alpha & \beta & \gamma & -T_1 \\ \alpha^2 & \beta^2 & \gamma^2 & -T_2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \tag{28}$$

In order for this homogenous system to be consistent with Eq.(25) the determinant of the matrix of coefficients must vanish.

$$\begin{vmatrix} 0 & 1 & 0 & -y \\ 1 & 1 & 1 & -T_0 \\ \alpha & \beta & \gamma & -T_1 \\ \alpha^2 & \beta^2 & \gamma^2 & -T_2 \end{vmatrix} \tag{29}$$

Since the determinant vanishes we can factor out the -1 from the last column and write the consistency condition as

$$\begin{vmatrix} 0 & 1 & 0 & y \\ 1 & 1 & 1 & T_0 \\ \alpha & \beta & \gamma & T_1 \\ \alpha^2 & \beta^2 & \gamma^2 & T_2 \end{vmatrix} = 0. \tag{30}$$

Multiplying the above determinant by the non-singular determinant

$$\begin{vmatrix} 1 & 1 & 1 & 0 \\ \alpha & \beta & \gamma & 0 \\ \alpha^2 & \beta^2 & \gamma^2 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix} \tag{31}$$

we find

$$\begin{vmatrix} 0 & 1 & 0 & y \\ 1 & 1 & 1 & T_0 \\ \alpha & \beta & \gamma & T_1 \\ \alpha^2 & \beta^2 & \gamma^2 & T_2 \end{vmatrix} \begin{vmatrix} 1 & \alpha & \alpha^2 & 0 \\ 1 & \beta & \beta^2 & 0 \\ 1 & \gamma & \gamma^2 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix} = \begin{vmatrix} 1 & \beta & \beta^2 & y \\ s_0 & s_1 & s_2 & T_0 \\ s_1 & s_2 & s_3 & T_1 \\ s_2 & s_3 & s_4 & T_2 \end{vmatrix} = 0, \tag{32}$$

where

$$s_i = \alpha^i + \beta^i + \gamma^i$$

and the s_i are symmetric functions in $\alpha, \beta,$ and γ.

Since the s_i's are sums of powers of α, β, γ and are symmetric functions, they can be expressed rationally in terms of the elementary symmetric functions. By expanding the final determinant in Eq. (32) along the top row and setting the expansion equal to zero to satisfy the consistency requirement we see that we have y expressed as a quadratic function of β along with the sums of powers of the three quantities α, β, γ. Summarizing the results obtained – we expand the above determinant along the top row and solve for y.

$$y = \frac{\begin{vmatrix} s_1 & s_2 & T_0 \\ s_2 & s_3 & T_1 \\ s_3 & s_4 & T_2 \end{vmatrix} - \beta \begin{vmatrix} s_0 & s_2 & T_0 \\ s_1 & s_3 & T_1 \\ s_2 & s_4 & T_2 \end{vmatrix} + \beta^2 \begin{vmatrix} s_0 & s_1 & T_0 \\ s_1 & s_2 & T_1 \\ s_2 & s_3 & T_2 \end{vmatrix}}{\begin{vmatrix} s_0 & s_1 & s_2 \\ s_1 & s_2 & s_3 \\ s_2 & s_3 & s_4 \end{vmatrix}}. \tag{33}$$

The right-hand-side depends quadratically on β and other expressions which are symmetric functions of $\alpha, \beta,$ and γ and hence demonstrate the claim made in Theorem (8.9).

The denominator of Eq. (33) is a function of the squares of the differences of $\alpha, \beta,$ and γ:

$$\triangle(\alpha, \beta, \gamma) = \begin{vmatrix} s_0 & s_1 & s_2 \\ s_1 & s_2 & s_3 \\ s_2 & s_3 & s_4 \end{vmatrix} = (\alpha - \beta)^2 (\alpha - \gamma)^2 (\beta - \gamma)^2 \tag{34}$$

Using this result we can write the solution for y as

$$\triangle(\alpha, \beta, \gamma) y = \begin{vmatrix} s_1 & s_2 & T_0 \\ s_2 & s_3 & T_1 \\ s_3 & s_4 & T_2 \end{vmatrix} - \beta \begin{vmatrix} s_0 & s_2 & T_0 \\ s_1 & s_3 & T_1 \\ s_2 & s_4 & T_2 \end{vmatrix} + \beta^2 \begin{vmatrix} s_0 & s_1 & T_0 \\ s_1 & s_2 & T_1 \\ s_2 & s_3 & T_2 \end{vmatrix}. \tag{35}$$

Returning to the original problem Eq. (23), we can similarly express the solution for ψ_j in terms of ϕ_j in the form of a determinant:

$$\begin{vmatrix} 1 & \phi_j & \phi_j^2 & \dots & \phi_j^{\rho-1} & \psi_j \\ s_0 & s_1 & s_2 & \dots & s_{\rho-1} & T_0 \\ s_1 & s_2 & s_3 & \dots & s_\rho & T_1 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ s_{\rho-1} & s_\rho & s_{\rho+2} & \dots & s_{2\rho-2} & T_{\rho-1} \end{vmatrix} = 0, \tag{36}$$

where $T_0, T_1, \ldots, T_{\rho-1}$ are all symmetric functions of $x_1, x_2, \ldots, x_n$ and $s_k = \phi_1^k + \phi_2^k + \phi_3^k + \ldots + \phi_\rho^k$.

The solution of these equations takes the form

$$\triangle(\phi_1, \phi_2, \phi_3, \ldots, \phi_\rho)\psi_j = A_0\phi_j^{\rho-1} + A_1\phi_j^{\rho-2} + \ldots + A_{\rho-1}, \tag{37}$$

where $\triangle$ is the product of the square of the differences of all the ϕ's (the square of the Vandermonde determinant) and $A_0, A_1, \ldots, A_{\rho-1}$ are all symmetric in $x_1, x_2, \ldots, x_n$.

Therefore ψ_j can be rationally expressed as a rational function of ϕ_j and the elementary symmetric functions of $x_1, x_2, \ldots, x_n$ (the coefficients of the polynomial that has $x_1, x_2, \ldots, x_n$ as roots).

Lagrange Interpolation Formula:
Given a set of $k+1$ nodes $\{x_0, x_1, \ldots, x_k\}$, which must all be distinct, $x_j \neq x_m$ for indices $j \neq m$, the *Lagrange basis* for polynomials of degree $\leq k$ for those nodes is the set of polynomials $\{l_0(x), l_1(x), \ldots, l_k(x)\}$ each of degree k which take values $l_j(x_m) = 0$ if $m \neq j$ and $l_j(x_j) = 1$. Using the Kronecker delta this can be written $l_k(x_m) = \delta_{jm}$. Each basis polynomial can be explicitly described by the product:

$$\begin{aligned} l_j(x) &= \frac{(x-x_0)}{(x_j-x_0)}\frac{(x-x_1)}{(x_j-x_1)}\cdots\frac{(x-x_{j-1})}{(x_j-x_{j-1})}\frac{(x-x_{j+1})}{(x_j-x_{j+1})}\cdots\frac{(x-x_k)}{(x_j-x_k)} \\ &= \prod_{\substack{0\leq m\leq k \\ m\neq j}} \frac{x-x_m}{x_j-x_m} \end{aligned}$$

Notice that the numerator $\prod_{m\neq j}(x-x_m)$ has k roots at the nodes $\{x_m\}_{m\neq k}$ while the denominator $\prod_{m\neq j}(x_j-x_m)$ scales the resulting polynomial so that $l_j(x_j) = 1$.

The Lagrange interpolating polynomial for those nodes through the corresponding values $\{y_0, y_1, \ldots, y_k\}$ is the linear combination

$$L(x) = \sum_{j=0}^{k} y_j l_j(x),$$

Each basis polynomial has degree k, so the sum $L(x)$ has degree $\leq k$, and it interpolates the data because

$$L(x_m) = \sum_{j=0}^{k} y_j l_j(x_m) = \sum_{j=0}^{k} y_j \delta_{mj} = y_m,$$

Theorem 8.10 (Extension of Lagrange's Theorem to Non Symmetric Functions) *If two rational functions of any set of variables are such that one remains unchanged by all the substitutions of the group to which the other belongs, the first is expressible by means of the second in the form of an integral polynomial whose coefficients are rational symmetric functions of the variables.*

Another statement of Lagrange's Theorem:

If a rational function $\phi(x_1, x_2, \ldots, x_n)$ remains unaltered by all the substitutions which leaves another rational function $\psi(x_1, x_2, \ldots, x_n)$ unaltered, then ϕ is a rational function of ψ and $c_1, c_2, \ldots, c_n$. Note that the group, G to which ϕ belongs contains the group, H to which ψ belongs, i.e. $H \subseteq G$. G is the wider group. Also ϕ is therefore the more symmetric function – a wider set of substitutions keep it unaltered.

Proof

$$\begin{array}{llll|r|r} I & h_2 & \ldots & h_P & \psi = \psi_1 & \phi = \phi_1 \\ g_2 & h_2 g_2 & \ldots & h_P g_2 & \psi_{g_2} = \psi_2 & \phi_{g_2} = \phi_2 \\ \ldots & \ldots & \ldots & \ldots & \ldots & \ldots \\ g_\nu & h_2 g_\nu & \ldots & h_P g_\nu & \psi_{g_\nu} = \psi_\nu & \phi_{g_\nu} = \phi_\nu \end{array} \tag{38}$$

The $\psi_1, \psi_2, \ldots, \psi_\nu$ are all distinct; but $\phi_1, \phi_2, ...\phi_n u$ need not be distinct since ϕ belongs to a group G which may be bigger than H. Under any substitution s on $x_1, x_2, \ldots, x_n$, the functions $\psi_1, \psi_2, \ldots, \psi_\nu$ are merely permuted. Moreover, if s replaces ψ_i by ψ_j, it replaces ϕ_i by ϕ_j. Set (using the Lagrange Interpolation formula)

$$\begin{aligned} g(t) &= (t-\psi_1)(t-\psi_2)\ldots,(t-\psi_\nu), \\ \lambda(t) &= g(t)\left(\frac{\phi_1}{t-\psi_1} + \frac{\phi_2}{t-\psi_2} + \ldots + \frac{\phi_\nu}{t-\psi_\nu}\right) \end{aligned}$$

so that $\lambda(t)$ is an integral function of degree $\nu - 1$ in t. Since $\lambda(t)$ remains unaltered under every substitution of s, its coefficients are rational symmetric functions of $x_1, x_2, \ldots, x_n$ and hence are rational functions of the elementary symmetric functions.

Now we're assuming here that $x_1, x_2, \ldots, x_n$ denote indeterminant quantities since $\psi_1, \psi_2, \ldots, \psi_\nu$ are algebraically distinct, so that $g'(\psi)$ is not idendically zero. In the case where special values are assigned to $x_1, x_2, \ldots, x_n$ such that two or more of the functions $\psi_1, \psi_2, \ldots, \psi_\nu$ become numerically equal, then $g'(\psi) = 0$, and ϕ is not a rational function of $\psi, c_1, ..., c_n$. Assuming that we're working with indetermiant $x_1, x_2, \ldots, x_n$, and taking $\psi_1 = \psi$ for t, we get

$$\lambda(\psi_1) = [(\psi_1 - \psi_2)(\psi_1 - \psi_3)\ldots(\psi_1 - \psi_\nu)]\,\phi_1$$

and in general

$$\phi = \frac{\lambda(\psi)}{g'(\psi)}.$$

Corollary 8.11 *A function can always be found in terms of which any number of given functions can be rationally expressed.*

The groups if the given functions have always one sub-group common to all, for the identity $S = 1$ is common to all. Therefore the functions can all be expressed in terms of any one of the functions peculiar to the common sub-group. Therefore if $\phi, \psi, \xi, \ldots,$ are the given functions, then

$$f = \alpha\phi + \beta\psi + \gamma\xi + \ldots,$$

where $\alpha, \beta, \gamma, \ldots,$ are arbitrary constants, is one such function for the common subgroup ; for any substitution which lease it unaltered must leave $\phi, \psi, \xi, \ldots,$ unaltered and must be common to the groups of $\phi_1, \psi_1, \xi_1, \ldots$.

Corollary 8.12 *Any rational function whatever can be rationally expressed in terms of a function having $n!$ distinct values ; in particular the Galois function.*

The Group of an $n!$-valued function, is included as a sub-group of every other permutation group.

Corollary 8.13 *The roots themselves can be expressed rationally in terms of the Galois function.*

Proof Let $f(x) = 0$ be the equation whose roots are $x_1, x_2, \ldots, x_n$, all supposed unequal; and let ψ_1 be a known value of a a rational function ψ of the roots which has $n!$ distinct values.

If all the roots except x_1 be permuted in every possible way, we obtain $\mu = (n-1)!$ distinct values of ψ given by the equation

$$F(x) = (\psi - \psi_1)(\psi - \psi_2) \ldots (\psi - \psi_\mu) = 0.$$

The coefficients of this equation when expanded are symmetric functions of $x_2, x_3, \ldots, x_n$ and therefore can be expressed in terms of the coefficients of

$$\frac{f(x)}{x - x_1} = 0,$$

and will involve x_1 in a rational form along with the coefficients of $f(x)$ If the expanded equation be represented by $F(\psi, x_1) = 0$, we have $F(\psi_1, x_1) = 0$, since it is satisfied by $\psi = \psi_1$; also we have $f(x_1) = 0$, from which it follows that the equation $f(x) = 0$ and $F(\psi_1, x) = 0$ have a common root. It is easily seen that this is the only common root. If therefore we seek the greatest common divisor of $f(x)$ and $F(\psi_1, x)$, as all the remainders vanish for $x = x_1$ in particular the remainder of the first degree in x of $f(x)$, in which the expression ψ_1 may be replaced by $\psi_2, \psi_3, \ldots \psi_\mu$ without altering its value.

Corollary 8.14 *All the values of the Galois function can be expressed rationally in terms of any one of them.*

Proof For all of these values belong to the same group, i.e., the identity.

9 Cauchy's 1815 Theorem

Cauchy's 1815 theorem says that if a function of n variables takes more than two distinct values when you permute the variables, then it must take at least p distinct values, where p is the largest prime less than or equal to n. The number of distinct values always divides n factorial. Symmetric functions take one value, and the only way to get exactly two are by use of the alternating functions, like the Vandermonde product.

Theorem 9.1 (Cauchy's theorem:) *For a function of n variables, $f = f(x_1, \ldots, x_n)$, the number of distinct values under all permutations of the variables is either 1, 2, or at least p, where p is the largest prime less than or equal to n.*

Proof Let f be a function of n (distinct) variables $f = f(x_1, \ldots, x_n)$. Permuting the variables produces up to $n!$ possible inputs, but we assume f to take only R distinct values under these permutations. By Lagrange's theorem $n!$ is divisible R. Assume for contradiction purposes that $2 < R < p$. Consider a p-cycle σ (a cyclic permutation of order p on p of the variables, fixing the rest). The sequence of function values obtained by applying successive powers of σ to a fixed initial ntuple of variables:

$$f(x),\ f(\sigma(x)),\ f(\sigma^2(x)),\ldots,\ f(\sigma^{p-1}(x)).$$

There are at most p distinct values in the above list (since σ^p is the identity). Let

$$v_k = f(\sigma^k(x)) \text{ for } k = 0, 1, ..., p-1,$$

stand for these p values and lest us apply the **Pigeonhole Principle** (i.e., if you have more pigeons than pigeonholes, at least one hole has more than one pigeon). These p values are drawn from the set of R assumed distinct possible values that f can ever take, and we assume $R < p$. These values satisfy the cyclic relation: $v_{k+1} = f(\sigma^{k+1}(x))$ is obtained from v_k by applying $\sigma \pmod p$. By the pigeonhole principle at least two of these p values must be equal.

$$v_r = v_s \text{ for some } 0 \le s < r \le p-1.$$

Let $d = r - s$ (so $1 \le d \le p-1$).

Now apply powers of σ repeatedly. Because σ shifts the indices cyclically (i.e., applying σ moves v_k to $v_{(k+1) \pmod p}$), the assumption $v_r = v_s$ means the sequence is periodic with a smaller step. The equality propagates under shifts: Apply σ any number of times to both sides of $v_r = v_s$. This gives: $v_{r+m} = v_{s+m \pmod p}$ for any integer m. In other words, shifting the indices by the same amount preserves the equality. The index differences are multiples of d: The indices where we know equalities hold are exactly the index positions that differ by multiples of $d \pmod p$. That is, for any integer m: $v_k = v_{(k+m-d) \pmod p}$, whenever the equality chain reaches there.

Primality of p makes d a generator: Crucially because p is prime and d is between 1 and $p-1$, we have $\gcd(d, p) = 1$. Therefore, d has a multiplicative inverse modulo p, and the multiples

$$\{0 \cdot d, 1 \cdot d, 2 \cdot d, \ldots, (p-1) \cdot d\} \pmod p$$

hit every residue class modulo p exactly once and we cycle through all residues modulo p. Starting from any fixed index k, we can reach any other index l by adding a suitable multiple of d modulo p. Therefore $v_k = v_l$ for all k, l. In group theory terms: the subgroup generated by d modulo p is the entire cyclic group $\mathbb{Z}/p\mathbb{Z}$. Therefore entire orbit of $f(\sigma^k x)$ for $k = 0, 1, 2, \ldots, p-1$ collapses to a single value.

Conclusion:

The shifts by multiples of d generate the entire cyclic group of order p. It follows that all the v_k are equal (the orbit collapses to a single value). In particular, $v_0 = v_1$, i.e., $f(x) = f(\sigma(x))$. Since the initial ntuple x was arbitrary, this holds for every starting assignment: σ leaves f invariant (does not change its value). By choosing different p-cycles (or by conjugacy/relabeling of variables), this applies to all p-cycles in S_n.

This implies that all p-cycles leave f invariant (fix every value). One can then construct a 3-cycle as a product (or commutator) of two such p-cycles. Thus, 3-cycles also fix values. A 3-cycle is a product of two transpositions. Analysis of transpositions then shows that either f is fully symmetric ($R = 1$) or it takes exactly two values (like the sign/alternating case, e.g., Vandermonde product). But we assumed $R > 2$, a contradiction. You can't get three or more without hitting at least p values. ∎

Thus, no such R with $2 < R < p$ is possible. The pigeonhole step forces p-cycles (and then smaller cycles) to preserve values, collapsing the possibilities to $R = 1$ or 2. This is the heart of Cauchy's 1815 argument (sometimes phrased in terms of"forms" or "arrangement" of the variables). Modern presentations often frame it in terms of the action of the symmetric group S_n on the values, with stabilizers and orbits (see the **Orbit-Stabilizer Theorem**) but the core pigeonhole idea on powers of a cycle remains the same.

Why Primality of p Is Essential
If p were composite, say d shares a common factor with p, the multiples of $d \pmod p$ would only generate a proper subgroup, and you might only force equality within cosets rather than full invariance under σ. The primality ensures the generated subgroup is the whole group, forcing full invariance. This is the precise bridge from the pigeonhole collision to invariance under the full p-cycle. Once you have invariance under all p-cycles, the rest of the proof proceeds by constructing smaller cycles (e.g., 3-cycles) as products/commutators and analyzing transpositions.

10 Wantzel's proofs

Theorem 10.1 *Every two valued function of n variables is of the form $S_1 \pm S_2\sqrt{\triangle}$, where S_1 and S_2 are integral symmetric functions, and $\triangle$ is the discriminant.*

Proof A two-valued function must belong the the group of order $\frac{1}{2}N$. The only group of this order is the alternating group to which the function $\sqrt{\triangle}$ belongs. The theorem therefore follows as an immediate consequence of the fundamental theorem Thm (8.9).

An Second Proof
Let two values of the function be denoted by ϕ_1 and ϕ_2 and let G_1 and G_2 be the corresponding groups, each of order $\frac{1}{2}N$. In the first place, these two groups must be identical; for it any substitution S of G_1 were to change ϕ_2 to its second value ϕ_1, then S^{-1} would change ϕ_1 into ϕ_2; but this is impossible, since S^{-1} and S each belong to G_1. Every substitution therefore of G_1 must belong to G_2 and vice-versa.

To show now that these groups coincided with the alternating group, consider the function $\psi = \phi_1 - \phi_2$. Any substitution which belongs to the common group leaves this unaltered; any other will change phi_1 to ϕ_2 and ϕ_2 to ϕ_1, and will therefore change the sign of ψ; some transposition, $(x_\alpha x_\beta)$ for example, will have this affect, for no group can include all the transpositions without coinciding with the symmetric group. It is easily inferred that $\psi_1 - \psi_2$ is divisible by $x_\alpha - x_\beta$, and hence by the product of all the differences, since ψ^2 is symmetric.

The quotient of ψ by $\sqrt{\triangle}$ is symmetric. To prove this, let $(\sqrt{\triangle})^m$ be the highest power of $\sqrt{\triangle}$ which occurs in ψ. The quotient of ψ by $(\sqrt{\triangle}^m$ is symmetric, since, if not, it would be an alternating function, and would again contain $\sqrt{\triangle}$ as a factor, which is contrary to assumption. It follows immediately that m is an odd number and that the quotient of ψ by $\sqrt{\triangle}$ is symmetric. Writing therefore $\psi_1 - \psi_2 = A\sqrt{\triangle}$ and $\phi_1 + \phi_2 = B$, where A and B are both symmetric, we have

$$\phi_1 = S_1 + S_2\sqrt{\triangle} \text{ and } \phi_2 = S_1 - S_2\sqrt{\triangle},$$

where S_1 and S_2 are both symmetric functions of the variables $x_1, x_2, \ldots, x_n$. The groups G_1 and G_2 obviously coincide with the group of $\sqrt{\triangle}$ – the Alternating Group.

Theorem 10.2 *Permutations are elements of the Symmetric Group and act on products of functions homomorphically because the action naturally distributes over multiplication. By definition, a permutation rearranges inputs, which respects function evaluation and standard pointwise multiplication. The following operations define the action of the Symmetric Group:*

$$\begin{aligned} \sigma \cdot (f+g) &= \sigma \cdot f + \sigma \cdot g; \\ \sigma \cdot (fg) &= (\sigma \cdot f)(\sigma \cdot g), \\ \tau \cdot (\sigma \cdot f) &= (\tau\sigma) \cdot f \end{aligned}$$

for $\sigma, \tau \in S_n$ and $f, g \in F[x_1, \ldots, x_n)]$. S_n is the set of permutations on n elements $[x_1, x_2, \ldots, x_n]$.

Proof 1. Define the Permutation Action A permutation σ acts on a function f by precomposition [11]:

$$(\sigma \cdot f)(x) = f(\sigma^{-1}(x))$$

2. Evaluate the Action on a Product of Functions We apply the permutation to the pointwise product of two functions, $(fg)(x) = f(x)g(x)$.

$$[\sigma \cdot (fg)](x) = (fg)(\sigma^{-1}(x)).$$

3. Unpack the Function Evaluation Evaluate the product at the permuted input $\sigma^{-1}(x)$:

$$[\sigma \cdot (fg)](x) = f(\sigma^{-1}(x)) \cdot g(\sigma^{-1}(x)).$$

4. Regroup using the Action Definition By the definition of the action in Step 1, we rewrite the scalar outputs:

$$[\sigma \cdot (fg)](x) = (\sigma \cdot f)(x) \cdot (\sigma \cdot g)(x).$$

5. Conclude for the Pointwise Product Since this holds for every element $x \in X$, the functions are equivalent:

$$\sigma \cdot (fg) = (\sigma \cdot f)(\sigma \cdot g).$$

Because this action distributes over the product of functions, it satisfies the required structural property (a homomorphism) for group actions on algebraic structures.

Theorem 10.3 *The alternating functions are the only unsymmetrical functions of n variables of which a power can be symmetric.*

Proof It is sufficient to prove this theorem for prime powers; for if there exists a function $F(x_1, x_2, \ldots, x_n)$ such that F^{pq} is symmetric and p is a prime, then there is also a function $\phi = F^q$ such that ϕ^p is symmetric.

Therefore let
$$\phi^p = S, \text{ a symmetric function.}$$
Since the group of ϕ which is unsymmetric, cannot contain all the transpositions, let $\sigma = (x_\alpha x_\beta)$ be a transposition which converts ϕ into ϕ_j; we have
$$\phi_j^p = \phi^p = S.$$
and therefore $\phi_j = \omega\phi$, where ω is a pth root of unity. Hence
$$\sigma\phi = \phi_j = \omega\phi,$$
and operating again with σ
$$\sigma^2\phi = \omega\sigma\phi = \omega^2\phi,$$
but $\sigma^2 = I$ and therefore $\omega^2 = 1$, and consequently $p = 2$.

Since ϕ^s is symmetric, ϕ is an alternating function. ∎

Theorem 10.4 *For any number, n, of independent elements there is no multiple-valied function of which a power is two-valiued when $n > 4$; and when $n = 3$, or $n = 4$, if there is any such power it is third power.*

Concentrating on prime numbers, and supposing that ϕ is a multiple-valued function whose p power is two-valued we have
$$\phi^p = S_1 + S_2\sqrt{\triangle}. \tag{39}$$
The group of ϕ cannot contain all the circular substitutions of the third order, for if it did, then this group would coincide with the alternating group, and ϕ would be two-valued. Let $\sigma = (x_\alpha, x_\beta, x_\gamma)$ be such a substitutions not contained in the group of ϕ and suppose $\sigma\phi = \phi_j$. From Eq. (39), since $S_1 + S_2\sqrt{\triangle}$ is unaltered by σ, we have
$$\phi^p = \phi_j^p,$$
hence $\phi_j = \omega\phi$, where ω is a p^{th} root of unity. Operating again twice in succession with σ we obtain

$$\begin{aligned}\sigma\phi &= \omega\phi,\\ \sigma^2\phi &= \omega\sigma\phi = \omega^2\phi,\\ \sigma^3\phi &= \omega^2\sigma\phi = \omega^3\phi,\end{aligned}$$

whence, since $\sigma^3 = 1$ (three cycle), we have $\omega^3 = 1$, and therefore $p = 3$.

Again, when the number of elements is greater than 4, there are circular substitutions of the fifth order, and those cannot all be contained in the group of ϕ (or else it will be two-valued and belong to the Alternating Group). Let τ be one of those not contained in this group and $\tau\phi = \phi_j$. We have, as before, from Eq. (39), by applying this substitution (which does not affect the right-hand side),

$$\phi^p = \phi_j^p = S_1 + S_2\sqrt{\triangle}.$$

Hence proceeding as before, we have $\tau\phi = \omega\phi$ and oeprating again on this and the succeeding equations with τ we easily find $\tau^5\phi = \omega^5\phi$; whence $\omega^5 = 1$, since $\tau^5 = 1$, and it is proved that $p = 5$. Now this results being incompatible with the previous value of p obtained, we infer that when the number of elements is greater than 4, it is impossile to find any multiple-valiued function ϕ, a prime power of which will be two-valued.

There are actually, when n is not greater than 4, multiple-valiued functions, a third power of which is two-valued.

11 Abel

After clearing the denominators to ensure all algebraic expressions are expressed as polynomials in the radicals, Niels Henrik Abel's 1826 proof proceeds through a structured combination of field extensions, permutation group properties, and a final proof by contradiction. Abel did not use modern Galois group vocabulary, but his strategy directly mapped out what we now call the Galois theory of fields.

1. The Structure of an Algebraic Solution Abel begins by formalizing what a general "solution by radicals" must look like. If a general quintic equation is solvable, its roots must be expressible by a finite chain of algebraic operations. He structures the general form of a root as a polynomial in a radical:$x = q_0 + q_1R^{1/m} + q_2R^{2/m} + \cdots + q_{m-1}R^{(m-1)/m}$ Here, m is a prime number, and the quantities q_i and R are either rational functions of the original coefficients, or they belong to a lower "order" field extension previously built up by adjoining other radicals.

2. Formulating Abel's Theorem on Radical Functions The core pivot of the proof relies on a fundamental theorem Abel establishes about the algebraic dependencies of these radicals. He proves that if a root of a general quintic can be written in the form above, then the radical itself, $R^{1/m}$, must be expressible as a rational function of the roots of the quintic and the coefficients. By applying permutations to the roots of the general quintic, Abel studies how the value of $R^{1/m}$ behaves. Because the coefficients of the general quintic are independent, independent variables, permuting the roots causes $R^{1/m}$ to cycle through its m distinct algebraic values (the m-th roots of unity multiplied by the radical).

3. Restricting the Prime Index m Abel analyzes what values the prime degree m of the outermost radical can take.A function of 5 independent variables can only take a specific number of distinct values when its variables are permuted. By invoking Cauchy's earlier work on the number of values a symmetric or partially symmetric function can take, Abel shows that the number of values a rational function of 5 variables can take under permutation must be a divisor of $5! = 120$. He narrows down the possible values of the prime index m. The only primes that can mathematically divide these algebraic combinations for 5 variables are $m = 2$ and $m = 5$.

4. Bounding the Value Multiplicity (The Cauchy Constraint) Abel then looks at a critical theorem proved by Augustin-Louis Cauchy: a non-symmetric function of 5 variables cannot take fewer than 5 distinct values unless it takes exactly 2 values (which corresponds to the alternating group A_5, generated by the square root of the discriminant). This creates a rigid algebraic trap:If $m = 5$, the radical expression must take exactly 5 distinct values under root permutations.Abel shows that the only way a function of 5 variables can take exactly 5 values under permutation is if it behaves like a specific linear combination of the roots (similar to a Lagrange resolvent).

5. The Final Contradiction Abel explicitly sets up the equations for these 5-valued functions. He assumes a root looks like:$x = q_0 + q_1 R^{1/5} + q_2 R^{2/5} + q_3 R^{3/5} + q_4 R^{4/5}$ By examining the algebraic behavior of the individual terms under 3-cycle and 5-cycle permutations, he derives a system of equations for the components q_i. He proves that for the general quintic, it is impossible for these equations to hold simultaneously. The permutations of the 5 independent roots require the individual components to vanish or collide in a way that contradicts the assumption that x is a root of a general equation with independent coefficients. Because the assumption that a general root can be expressed as a chain of radicals leads directly to an algebraic impossibility, Abel concludes that the general quintic equation cannot be solved by radicals.

Summary of Abel's Path After Clearing Denominators

Defines Root Anatomy: Expresses the hypothetical root as a polynomial of a prime-degree radical $R^{1/m}$.

Inverts the Radical: Proves that $R^{1/m}$ must be a rational function of the quintic's roots.

Applies Permutations: Analyzes how $R^{1/m}$ changes values when the roots are permuted.

Invokes Cauchy's Bound: Restricts the possible values of m to 2 or 5 based on the combinatorics of 5 variables.

Forces Contradiction: Demonstrates that the algebraic conditions required to form a 5-valued function cannot be met by the coefficients of a general quintic.

To understand how Augustin-Louis Cauchy's work on permutations constrained Niels Henrik Abel, we have to look at the combinatorics of switching variables.
When Abel proved that the outermost radical of a hypothetical solution, $v = R^{1/m}$, must be a rational function of the five roots $(x_1, x_2, x_3, x_4, x_5)$, he created a bridge between algebra and combinatorics. Because the coefficients of a general quintic are completely independent, permuting the roots creates a set of distinct values for that function.
Cauchy's theorem placed a strict mathematical ceiling on exactly how many values such a function could take, narrowing Abel's options down to a dead end.

1. The Core of Cauchy's Theorem (1815) In 1815, Cauchy published a groundbreaking theorem concerning symmetric and non-symmetric functions. He proved a rigid restriction for functions of n independent variables

Theorem 11.1 (Cauchy's Number of Values Theorem:) *If a non-symmetric function of n independent variables changes its value when the variables are permuted, the number of distinct values it can take (its index) must be at least the largest prime factor of n. Furthermore, for $n = 5$, a function cannot have 3 or 4 distinct values. It can only have 1 (fully symmetric), 2, 5, or more (up to $5! = 120$) distinct values.*

2. How This Restricted Abel's Prime Radical (m) Abel assumed the general root of the quintic could be written as a polynomial in a radical of prime degree m:$x = q_0 + q_1 v + q_2 v^2 + \cdots + q_{m-1} v^{m-1}$ where $v = R^{1/m}$ By definition, if you multiply v by the m-th root of unity (ω), its m-th power R stays exactly the same. Therefore, the function $v(x_1, x_2, x_3, x_4, x_5)$ can take exactly m distinct values as its variables are permuted. Abel applied Cauchy's theorem to this m:Because v takes exactly m values, Cauchy's theorem dictates that m must be a possible number of values for a 5-variable function. According to Cauchy, for 5 variables, the number of values can only be 1, 2, 5, or ≥ 6. Since m must be a prime number by algebraic design, the only prime numbers allowed in that set are $m = 2$ and $m = 5$. Cauchy's theorem instantly destroyed the possibility of using cubic roots ($m = 3$), seventh roots ($m = 7$), or any higher prime radicals at the final stage of a quintic solution.

3. Eliminating the $m = 2$ Option An $m = 2$ radical means a square root. If $m = 2$, the function v takes exactly 2 values under permutation. Cauchy's work had already thoroughly mapped out 2-valued functions of 5 variables. The only way a function of 5 variables can take exactly 2 values is if it is a symmetric function multiplied by the alternating sign function (the square root of the discriminant, $\sqrt{\Delta}$). Abel knew that adjoining a square root ($\sqrt{\Delta}$) merely splits the symmetric group S_5 into the alternating group A_5. It simplifies the problem slightly but does not solve the equation. A square root alone cannot express five distinct, independent roots. Therefore, $m = 2$ could not be the final step.

4. Trapping the $m = 5$ Option This left Abel with exactly one remaining choice: $m = 5$. The outermost radical had to be a fifth root, meaning the function v must take exactly 5 distinct values when the roots are permuted. Abel turned back to Cauchy's structural analysis of 5-valued functions. Cauchy had proven that if a function of 5 variables has exactly 5 values, the permutations that leave the function unchanged must form a specific subgroup of order 24 (isomorphic to the symmetric group S_4). To satisfy this rigid combinatorial structure, Abel showed that the function v must be a linear combination of the roots structured precisely like a Lagrange resolvent:$v = x_1 + \alpha x_2 + \alpha^2 x_3 + \alpha^3 x_4 + \alpha^4 x_5$(where α is a fifth root of unity).

5. The Closing of the Trap By using Cauchy's theorem to force v into this exact, unyielding algebraic shape, Abel was able to execute his final blow. He applied a simple 3-cycle permutation (e.g., swapping $x_1 \to x_2 \to x_3 \to x_1$) to the roots. Because a 3-cycle permutation belongs to the alternating group A_5 but not to the subgroup that keeps a 5-valued function stable, it forced a contradiction. Abel proved that under these required permutations, the algebraic coefficients would have to simultaneously equal completely different values, meaning the assumed fifth-root function could not actually exist for independent variables. Without Cauchy's theorem bounding the permissible number of values to 2 and 5, Abel would have had to fight an infinite number of potential radical degrees ($m = 3, 5, 7, 11\ldots$). Cauchy provided the boundaries that allowed Abel to corner the quintic and prove it impossible.

In his 1826 paper, Niels Henrik Abel used a beautifully simple property of 3-cycle permutations to snap his algebraic trap shut. He didn't use modern group theory notation (like A_5 or S_5), but he expressed the exact same logical mechanism. Here is the exact step-by-step breakdown of how a single 3-cycle permutation created his final, fatal contradiction.

1. The Setup: The Five-Valued RadicalThrough Cauchy's theorem, Abel established that the final, outermost radical needed to solve the quintic must be a fifth root, which we can call v:$v = R^{1/5}$ Because v is a fifth root, permuting the roots of the quintic can only change v by multiplying it by some fifth root of unity, ω (where $\omega^5 = 1$). Therefore, under the full group of permutations, v can take exactly 5 distinct values:$\{v, \omega v, \omega^2 v, \omega^3 v, \omega^4 v\}$

2. Applying the 3-Cycle Permutation Abel then chose an arbitrary 3-cycle permutation of the roots, which we can denote as p. A 3-cycle simply takes three roots and rotates them in a loop (for example, swapping $x_1 \to x_2 \to x_3 \to x_1$), leaving the other two roots completely untouched. Let's apply this 3-cycle p to our radical function v. As established, applying any permutation to the roots can only multiply v by some fifth root of unity, say ω^k:$p(v) = \omega^k \cdot v$

3. The Power of Order Three A 3-cycle has a strict geometric property: if you execute it exactly three times, every element returns to its original starting position ($p^3 = e$, the identity permutation). Abel applied the exact same 3-cycle permutation three times in a row to both sides of the equation:

First time: $p(v) = \omega^k \cdot v$

Second time: $p(p(v)) = \omega^k \cdot \omega^k \cdot v = \omega^{2k} \cdot v$

Third time: $p(p(p(v))) = \omega^k \cdot \omega^k \cdot \omega^k \cdot v = \omega^{3k} \cdot v$

Because applying a 3-cycle three times restores the roots to their original order, the left side must evaluate back to the original, un-permuted value of v:$v = \omega^{3k} \cdot v$ For this equation to hold true, the multiplier must equal 1:$\omega^{3k} = 1$

4. The Arithmetic Lock We now have two strict rules dictating the behavior of this multiplier:From the nature of the fifth root, ω is a fifth root of unity ($\omega^5 = 1$). From the nature of the 3-cycle, ω^{3k} must equal 1. Because 3 and 5 are prime numbers that share no common factors (coprime), the only power that satisfies both conditions simultaneously is $k = 0$, meaning $\omega^0 = 1$. Therefore:$p(v) = 1 \cdot v = v$ This reveals an astonishing fact: every single 3-cycle permutation leaves the radical v completely unchanged.

5. The Final Contradiction This is where Abel applies the coup de gräce using Cauchy's framework:

> Fact A: In combinatorics, the 3-cycle permutations are the foundational building blocks that generate the entire **Alternating Group** (A_5). Because v is unaffected by every 3-cycle, it is completely invariant under the entire alternating group.

Fact B: According to Cauchy's structural theorems, any function of 5 variables that is completely invariant under the alternating group can take **at most 2 distinct values** across all possible permutations (acting like the square root of a discriminant).

Now, look at the two numbers we have forced upon the exact same function v: From its definition as a fifth root, v must take exactly 5 values. From the 3-cycle arithmetic, v can take at most 2 values. A function cannot simultaneously possess exactly 5 values and at most 2 values. The assumption that a general quintic can be solved by a chain of radicals containing a fifth root collapses into an absolute algebraic impossibility.

12 The Commutator Subgroup (The Abelian Quotient)

Definition Let G be a group. An element $g \in G$ is called a commutator if $g = aba^{-1}b^{-1}$ for elements $a, b \in G$.
The smallest subgroup that contains all commutators of G is called the commutator subgroup or derived subgroup of G, and is denoted by G'.

A highly practical relationship emerges with the commutator subgroup G'. The quotient G/G' forms the largest Abelian (commutative) group that can be created from G. If you "mod out" by the commutator subgroup, all order-dependent operations are eliminated.

Proposition 12.1 *Let G be a group with commutator subgroup G'.*

(a) The subgroup G' is normal in G, and the factor group G/G' is Abelian.

(b) If N is any normal subgroup of G, then the factor group G/N is Abelian, if and only if $G' \in N$.

Proof (a) Let $x \in G'$ and let $g \in G$. Then $gxg^{-1}x^{-1} \in G'$, and since $x \in G'$, we must have $gxg^{-1} = gxg^{-1}x^{-1}x \in G'$.

The factor group G/G' must be Abelian since for any cosets aG', bG', because G' is normal, we have $xG' = G'x$ for all $x \in G$.

$$aG'bG'a^{-1}G'b^{-1}G' = abG'G'a^{-1}b^{-1}G'G' = aba^{-1}b^{-1}G' = G'.$$

Proof (b) Let N be a normal subgroup of G. If $N \supseteq G'$, then G/N is a homomorphic image of G/G' and must be Abelian. Conversely, suppose that G/N is Abelian. Then $aNbN = bNaN$ for all $a, b \in G$, or simply $aba^{-1}b^{-1}N = N$, showing that every commutator of G belongs to N. This implies $G' \subseteq N$. ∎

Theorem 12.2 *The Symmetric Group G on n letters is not solvable unless $n \leq 4$.*

Proof Let $G = S_0 \supseteq S_1 \supseteq S_2 \supseteq \ldots \supseteq S_s$ be any chain of subgroups, each normal in the preceeding with cyclic quotient-group S_{k-1}/S_k; we shall prove by induction on s that S_s must contain every 3-cycle (ijk). This will imply that $S_s > I$, and so, that G cannot be solvable. Since $S_0 = G$ contains every 3-cycle, it is sufficient to show that if S_{k-1} contains every 3-cycle, then so does S_k. First note that if the permutations ϕ and ψ are both in S_{k-1}, then their so-called commutator $\gamma = \phi^{-1}\psi^{-1}\phi\psi$ is in S_s. To see this, consider their images ϕ' and ψ' in S_{k-1}/S_k. The latter, being cyclic, is commutative; hence

$$\gamma' = \phi'^{-1}\psi'^{-1}\phi'\psi' = \phi'^{-1}\psi'^{-1}\psi'\phi' = I \text{ in } S_{k-1}/S_k,$$

which implies that $\gamma \in S_k$. But in the special case when $\phi = (ijk)$ and $\psi = (jkm)$, where i, j, k are given and l, m are any two other letters (such letters exist unless $n \leq 4$), we have

$$\gamma = (jli)(mkj)(ilk)(jkm) = (ijk) \in S_k, \text{ for all } i, j, k.$$

This proves that S_k contains every 3-cycle, as desired.

The quotient chain doesn't terminate on the Identity, therefore the group G is not solvable unless $n \leq 4$. The 3-cycles generate the Alternating group which is simple and has no normal subgroups.

13 Hamilton

In 1836, the brilliant Irish mathematician Sir William Rowan Hamilton read Niels Henrik Abel's 1826 paper and found parts of the proof to be brilliant but "obscure" and "problematic". Recognizing gaps in how Abel handled permutations and nested radical operations, Hamilton published a massive, highly meticulous 130-page refinement in 1839 entitled "On the Argument of Abel". Hamilton's core contribution was cleaning up the 3-cycle permutation logic. He realized that Abel had jumped to conclusions regarding how nested radicals behave when root transformations occur. Hamilton rebuilt the 3-cycle argument into a completely bulletproof algebraic mechanism.

1. Identifying Abel's Flaw: The Hidden "Ambiguity" Abel had asserted that because a radical $v = R^{1/5}$ must take 5 distinct values under permutation, applying a 3-cycle permutation must multiply it by some fifth root of unity ω^k. Hamilton pointed out a deep issue here: radicals are multi-valued expressions. When you apply a permutation to the roots of a polynomial, you alter the underlying variables. You cannot simply assume that the radical function tracks to the exact same branch of the root without a rigorous tracking mechanism. Hamilton realized that if the radical inside the function was nested under other radicals, a permutation might swap the branches of the inner radicals, destroying Abel's straightforward equation $p(v) = \omega^k v$.

2. Hamilton's Fix: Commutators of 3-Cycles To resolve this branch-switching ambiguity, Hamilton bypassed simple 3-cycles and introduced what modern group theorists call commutators. Instead of just checking what a single 3-cycle does to a radical, Hamilton looked at the interaction between two different 3-cycles, which we can call P and Q. He analyzed the specific composite operation:$\text{Commutator} = P \cdot Q \cdot P^{-1} \cdot Q^{-1}$ Hamilton proved a stunning combinatorial property of five variables: you can construct any 3-cycle permutation of 5 elements by combining the commutators of other 3-cycles. For example, let:P be the 3-cycle swapping roots $(1, 2, 3)Q$ be the 3-cycle swapping roots $(3, 4, 5)$ If you apply them in the sequence $P \to Q \to P^{-1} \to Q^{-1}$, the net result is a completely new 3-cycle permutation.

3. Neutralizing the Branch-Switching Problem Why did Hamilton use this complex, four-step sequence instead of Abel's single 3-cycle? Because the commutator structure cancels out any branch-switching ambiguities of nested radicals. Let's look at how Hamilton applied this to the 5-valued radical v: Suppose permutation P multiplies v by some root of unity and shifts an inner branch. Suppose permutation Q multiplies v by another root of unity and shifts an inner branch differently. When you run the sequence in reverse (P^{-1} then Q^{-1}), the operations on the internal branches perfectly unwind and cancel each other out, because the underlying field extensions are Abelian. However, the fifth root of unity multipliers are just simple complex numbers, which are commutative ($\omega^a \cdot \omega^b = \omega^b \cdot \omega^a$). Therefore, the multipliers track through the commutator like this:

$$v \xrightarrow{P} \omega^a v \xrightarrow{Q} \omega^b \omega^a v \xrightarrow{P^{-1}} \omega^{-a} \omega^b \omega^a v \xrightarrow{Q^{-1}} \omega^{-b} \omega^{-a} \omega^b \omega^a v$$

Because complex multiplication commutes, the powers completely annihilate one another:

$$\omega^{-b} \omega^{-a} \omega^b \omega^a = \omega^{-b-a+b+a} = \omega^0 = 1$$

4. The Refined Contradiction Hamilton's math forced a beautiful trap:

By Combinatorics: The commutator sequence $PQP^{-1}Q^{-1}$ successfully creates a brand new, non-trivial 3-cycle permutation that alters the roots.

By Algebra: The exact same commutator sequence reduces the radical's multiplier to exactly 1, meaning the radical v is left entirely unchanged.

Hamilton proved that every single 3-cycle permutation must leave the value of the radical v completely invariant, regardless of how many nested, lower-order radicals were sitting underneath it. This completely sealed the mathematical loophole in Abel's paper. Since the 3-cycles generate the alternating group A_5, the radical v cannot possess 5 distinct values, and the proof of the quintic's insolvability was officially completed with total rigor.

Table 2: **Comparison of the 3-Cycle Logic**

Feature	**Abel's Original Argument (1826)**	**Hamilton's Refinement (1839)**
Primary Tool	Compact: A single 3-cycle permutation P	A commutator of two 3-cycles ($PQP^{-1}Q^{-1}$)
Handling of Nested Roots	Assumed inner radicals wouldn't interfere with the multiplier	Explicitly proved that inner radical branch-shifts cancel out
Mathematical Rigor	Had minor logical gaps regarding multi-valued functions	Completely seamless: closed all algebraic loopholes
Modern Equivalent	Showing that A_5 acts trivially on radicals	Proving that the commutator subgroup of A_5 is A_5 itself (i.e., A_5 is a perfect group)

To map out the algebraic field extensions with total precision and prevent any hidden multi-valued ambiguity, Sir William Rowan Hamilton developed an incredibly strict, highly formalized notation system in his 1839 paper, "On the Argument of Abel". Instead of writing standard radical signs like $\sqrt{x}$ or $\sqrt[5]{R}$ – which he argued were mathematically dangerous because they obscured multi-valued branch-switching – Hamilton used nested levels of accent marks (!, !!, !!!) appended to the variable names, combined with explicit prime indexes.

1. The Base Fields and Independent Variables Hamilton starts with the arbitrary coefficients or roots of the general n-th degree polynomial. He denotes these completely independent variables using basic lowercase subscripts: $a_1, a_2, a_3, \ldots, a_n$ Any function that can be built by simple addition, subtraction, multiplication, or division of these variables is written as a standard rational function:$f_1(a_1, a_2, \ldots, a_n)$

2. First-Order Radicals (The Single Accent !) When the first layer of radicals is adjoined to the base field (e.g., square roots or cube roots of the initial coefficients), Hamilton marks them with a single exclamation point accent (!). He defines a set of n' distinct first-order radicals $a'_1, a'_2, \ldots, a'_{n'}$ by their explicit algebraic relations:

$$\begin{aligned} (a'_1)^{\alpha'_1} &= f_1(a_1, a_2, \ldots, a_n) \\ (a'_2)^{\alpha'_2} &= f_2(a_1, a_2, \ldots, a_n) \\ &\vdots \\ (a'_{n'})^{\alpha'_{n'}} &= f_{n'}(a_1, a_2, \ldots, a_n) \end{aligned}$$

Meaning: α'_i is the prime root index (like 2, 3, or 5). The variable a'_1 represents the radical itself, structurally tied to a specific base rational function.

3. Second-Order Radicals (The Double Accent !!) To build the next layer of the tower (radicals containing other radicals inside them), Hamilton adds a second exclamation point (!!). These new radicals are functions of both the base variables and the newly created first-order radicals:

$$(a''_1)^{\alpha''_1} = f'_1(a'_1, a'_2, \ldots, a'_{n'}, a_1, a_2, \ldots, a_n)$$

By forcing the notation to explicitly list every single sub-radical parameter in the parentheses, Hamilton guaranteed that if a permutation altered a lower-tier root (like a'_1), its exact ripple effect onto the higher-tier root (a''_1) could be meticulously traced.

4. How Hamilton Explicitly Wrote a Cubic Root Solution To show how his notation mapped to historical solutions before attacking the quintic, Hamilton explicitly wrote out the classical Cardano solution for the general cubic equation

$$x^3 + a_1 x^2 + a_2 x + a_3 = 0$$

using his system: He tracked the solution through a strict sequence of rational functions (c_1, c_2) and nested radical levels (a'_1, a''_1):

Symmetric Base Functions:

$$\begin{aligned} c_1 &= -\frac{1}{54}(2a_1^3 - 9a_1 a_2 + 27a_3) \\ c_2 &= \frac{1}{9}(a_1^2 - 3a_2) \end{aligned}$$

First-Order Radical (Level 1): A square root of the discriminant components:

$$(a'_1)^2 = f_1(a_1, a_2, a_3) = c_1^2 - c_2^3$$

Second-Order Radical (Level 2): A cube root that nests the first square root inside it:

$$(a_1'')^3 = f_1'(a_1', a_1, a_2, a_3) = c_1 + a_1'$$

Finally, he expresses the absolute root x as a pure, single-valued algebraic function b:$x = b = f(a_1'', a_1', a_1, a_2, a_3) = -\frac{a_1}{3} + a_1'' + \frac{c_2}{a_1''}$

Why this Notation Settled the Argument Whenever previous mathematicians (including Ruffini) tried to permute the variables in a radical expression, critics argued that equations like $\sqrt{R} \to -\sqrt{R}$ were ambiguous because $\sqrt{R}$ didn't have an inherent, single-valued identity. By replacing standard radical signs with explicitly defined variables (a_1', a_1'') indexed to precise functional domains, Hamilton turned multi-valued radical expressions into a precise, deterministic system of tracking algebraic field elements.

A modern exposition of Hamilton's proof modern exposition translates his cumbersome, 130-page tower of algebraic notation into the streamlined language of modern abstract algebra, field extensions, and topological/geometric tracking. Historically, this presentation bridges the gap between Abel's pure algebraic manipulation and Galois's abstract group theory. A famous contemporary variant of this approach was popularized by the legendary mathematician Vladimir Arnold, who geometricized Hamilton's permutation logic [10].

1. The Setup: Towers of Radical Extensions In modern field theory, solving an equation by radicals means constructing a sequential chain of field extensions (a radical tower). We start with the base field $F_0 = \mathbb{C}(a_1, a_2, \ldots, a_5)$ generated by the independent coefficients of the quintic. A solution implies reaching a final field F_k that contains the roots, where each step in the tower adjoins a prime-degree radical:

$$F_0 \subset F_1 \subset F_2 \subset \cdots \subset F_k$$

Where $F_{i+1} = F_i(\sqrt[m_i]{R_i})$ for some $R_i \in F_i$, and m_i is a prime number.

2. The Tracking Mechanism (Hamilton's Riemann Surfaces) In a modern topological or complex-analytic presentation, we treat the roots $x_1, \ldots, x_5$ as multi-valued continuous functions mapping from the space of coefficients. When we vary the coefficients along a closed loop in complex space, the coefficients return to their exact starting values. However, the roots $x_1, \ldots, x_5$ trace paths that may permute their identities (e.g., executing a 3-cycle like $x_1 \to x_2 \to x_3 \to x_1$). Hamilton's foundational insight was figuring out how the radical tower handles these root loops.

3. The Commutator Loop Let P and Q be two distinct loops in the coefficient space that cause 3-cycle permutations on the roots:Loop P induces the permutation $\sigma = (1\,2\,3)$Loop Q induces the permutation $\tau = (3\,4\,5)$ Now, consider a composite loop known as the commutator loop:$\gamma = P \cdot Q \cdot P^{-1} \cdot Q^{-1}$ If you track the permutation of the roots along the commutator loop γ, the net result is a brand-new 3-cycle permutation: $\gamma \implies \sigma\tau\sigma^{-1}\tau^{-1} = (1\,2\,3)(3\,4\,5)(1\,3\,2)(5\,4\,3) = (1\,3\,4)$

4. The Algebraic Triviality of the Loop Let's look at what happens to the elements inside the radical tower when we travel along this commutator loop γ. The first extension is $F_1 = F_0(\sqrt[m]{R_0})$, where

R_0 is a basic rational function of the coefficients. Traveling along any closed loop P or Q can only multiply the radical $v = \sqrt[m]{R_0}$ by some root of unity ω^a or ω^b.

Because complex numbers commute ($\omega^a \cdot \omega^b = \omega^b \cdot \omega^a$), tracking the value of v along the commutator loop $\gamma = PQP^{-1}Q^{-1}$ results in: $\omega^a \cdot \omega^b \cdot \omega^{-a} \cdot \omega^{-b} = \omega^{a+b-a-b} = \omega^0 = 1$. Therefore, traveling along the commutator loop completely un-twists the radical and forces it back to its original branch value. The function v is invariant under γ.

5. Climbing the Tower Hamilton's absolute refinement over Abel was proving that this invariance propagates perfectly up every single level of the tower.

> Suppose every radical up to field F_i returns to its starting value after running a commutator loop of commutators.
>
> Then any new radical adjoined in F_{i+1} will also see its branch-shifts commute and cancel out if the loop is nested deep enough.

In modern group theory language, we are exploiting the fact that the Symmetric Group (S_5) and the Alternating Group (A_5) are perfect groups. Their commutator subgroups do not shrink:

$$[A_5, A_5] = A_5$$

No matter how many times you take commutators of commutators, you can always find a loop sequence that generates a non-trivial 3-cycle permutation on the roots.

6. The Modern Contradiction

> From the Geometry/Topology: We can construct a highly nested commutator loop $\gamma = [[P, Q], [R, S]] \dots$ that still yields a non-trivial 3-cycle permutation on the roots, meaning the roots **must change position**.
>
> From the Algebra: Because the loop is a deeply nested commutator, every single layer of the radical tower is forced to evaluate cleanly back to a multiplier of 1, meaning any formula built of radicals **cannot change value**.

Because a valid formula for a root must change when the root itself changes, the radical formula cannot exist.

Key Takeaway The modern view of Hamilton's proof shows that algebraic solvability requires a group that can be broken down into trivial steps via commutators (a solvable group). Because the root permutations of a quintic (A_5) constantly regenerate themselves under commutator operations, they create an infinite geometric loop that any finite tower of radicals is structurally incapable of untangling.
To understand why a highly nested commutator loop of 3-cycles never shrinks or vanishes, we have to look at the intersection of topology and group theory. This is the exact mechanism that Vladimir Arnold geometricized in his modern exposition of Abel and Hamilton's work.

The core secret lies in a unique combinatorial property of 5 or more items: the alternating group A_5 is a "perfect group," meaning it is its own commutator subgroup.

1. The Geometric Setup: Loops as Permutations Imagine the five roots of the quintic as five distinct points on a complex plane.

A loop in the space of coefficients acts like a smooth animation that moves these five points around each other before bringing them back to the exact same five starting slots.

Even though the slots are the same, individual roots have swapped places. Therefore, every continuous loop maps directly to a specific permutation.

A loop that swaps three roots in a circle maps to a 3-cycle.

2. The Mechanics of a Base Commutator $[P, Q]$ Let's see what happens geometrically when we build a simple commutator loop from two different 3-cycle loops, P and Q. Suppose:

Loop P moves the roots in slots 1, 2, and 3. Its permutation is $\sigma = (1\,2\,3)$.

Loop Q moves the roots in slots 3, 4, and 5. Its permutation is $\tau = (3\,4\,5)$.

Notice that these loops overlap at slot 3. This overlap is where the geometric magic happens. Let's trace a physical point starting in slot 1 through the commutator loop $\gamma = P \cdot Q \cdot P^{-1} \cdot Q^{-1}$:

Run Loop P: The root in slot 1 moves to slot 2.

Run Loop Q: Loop Q only affects slots 3, 4, and 5. Our root (now in slot 2) sits completely still.

Run Loop P^{-1} (Reverse of P): This moves whatever is in slot 2 back to slot 3. Our root is now in slot 3.

Run Loop Q^{-1} (Reverse of Q): Loop Q^{-1} rotates slots $(5\,4\,3)$ into $(3\,5\,4)$. Therefore, the root sitting in slot 3 gets pushed into slot 4.

By tracking every slot this way, the net result of running this four-step loop sequence is:

$$\sigma\tau\sigma^{-1}\tau^{-1} = (1\,2\,3)(3\,4\,5)(1\,3\,2)(5\,4\,3) = (1\,3\,4)$$

The Geometric Upshot: By combining four loops that only perform 3-cycles, we did not unwind the permutation. Instead, we generated a brand-new, completely non-trivial 3-cycle permutation $(1\,3\,4)$.

3. Climbing the Nesting Ladder: $[[P, Q], [R, S]]$ Because the commutator of two 3-cycles yields another 3-cycle, we can repeat this process infinitely. Let's define two separate commutator loops, each engineered from overlapping 3-cycles:

Let loop $A = [P, Q]$, which yields the 3-cycle $\alpha = (1\,3\,4)$.

Let loop $B = [R, S]$, which yields the 3-cycle $\beta = (2\,4\,5)$.

Now, we nest them into a deeper commutator loop: $\gamma_{\text{nested}} = [A, B] = A \cdot B \cdot A^{-1} \cdot B^{-1}$ Because A and B are both 3-cycles that overlap at slot 4, running this highly nested loop sequence will, by the exact same algebraic logic, yield another brand-new 3-cycle: $\alpha\beta\alpha^{-1}\beta^{-1} = (1\,3\,4)(2\,4\,5)(1\,4\,3)(5\,4\,2) = (1\,4\,2)$ Even though γ_{nested} is a massive string of 16 individual base loops wrapped inside each other, **the roots are still actively swapping places at the end of the set of permurations**.

4. Why This Fails for Lower-Degree Equations (The Contrast) To see why this topological property traps the quintic, look at what happens if you try this with a cubic equation (3 roots).

The only 3-cycles available for 3 roots are $(1\,2\,3)$ and $(1\,3\,2)$.

If you compute their commutator:$(1\,2\,3)(1\,3\,2)(1\,3\,2)(1\,2\,3) = \text{identity (no movement)}$

For 3 roots (and similarly for 4 roots), if you nest commutators deep enough, the permutations eventually crash into the identity. The loops completely untangle, the roots stop moving, and the contradiction vanishes –which is precisely why cubics and quartics can be solved by radicals.

5. The Trap Closes For 5 or more roots, you can nest commutators 100 levels deep, and you will never run out of overlapping 3-cycles. You can always find a loop sequence where the roots physically swap positions at the end.

Yet, as Hamilton proved, a 100-level-deep nested commutator forces the algebra of any radical formula to cancel its multipliers out to exactly 1, meaning the formula outputs the exact same value it started with.

The topology proves the roots **must move**, while the algebra proves the radical formula **cannot move**. This permanent geometric mismatch is the ultimate proof that a general formula for the quintic cannot exist.

Nesting commutators deepens the loop chain to **force each successive layer of the radical tower to evaluate to a multiplier of 1, effectively neutralizing them one by one**.

In abstract algebra, we call this climbing down the derived series of a group. Geometrically, each level of nesting strips away the multi-valued "twisting" of a specific tier of radicals, exposing the next layer down until the entire formula is forced to act like a single-valued rational function.

1. The Anatomy of a Tiered Radical Tower To see how nesting targets specific levels, we must look at how an algebraic formula is built. Every radical solution is a nested hierarchy of dependencies:

Level 1 Radicals (F_1): Radicals whose inputs are simple, single-valued rational functions of the coefficients (e.g., $v_1 = \sqrt[5]{R_0}$).

Level 2 Radicals (F_2): Radicals whose inputs contain Level 1 radicals inside them (e.g., $v_2 = \sqrt[3]{5 + v_1}$).

Level 3 Radicals (F_3): Radicals whose inputs contain Level 2 radicals (e.g., $v_3 = \sqrt[2]{\cdots + v_2}$).

When you move the coefficients of a polynomial along a single closed loop, **every level of radical can twist simultaneously**. A single loop might multiply v_1 by ω, which shifts the input of v_2, causing v_2 to jump to a completely different Riemann sheet. It looks like chaotic, untrackable behavior.

2. A 1-Level Commutator $[P, Q]$ Targets Level 1 Radicals Suppose we run a simple commutator loop $\gamma_1 = PQP^{-1}Q^{-1}$.

Let's look at a Level 1 radical, $v_1 = \sqrt[m]{R_0}$. Because R_0 is just a basic rational function of the coefficients, it has no multi-valued ambiguity. Loop P can only multiply v_1 by some constant complex root of unity ω^a, and loop Q multiplies it by ω^b.

Because complex numbers commute, tracking v_1 along the 4-step sequence γ_1 gives:$v_1 \to \omega^a v_1 \to \omega^b \omega^a v_1 \to \omega^{-a} \omega^b \omega^a v_1 \to \omega^{-b} \omega^{-a} \omega^b \omega^a v_1 = 1 \cdot v_1$

The Result: The 1-level commutator has completely neutralized Level 1 radicals. After running γ_1, v_1 returns exactly to its starting branch. To Level 1 radicals, the loop γ_1 acts exactly like the identity loop (doing nothing).

3. A 2-Level Nested Commutation $[A, B]$ Targets Level 2 Radicals Now, what happens to a Level 2 radical, $v_2 = \sqrt[k]{f(v_1)}$, when we run that same 1-level commutator γ_1?

Even though v_1 returns to its original value at the end of γ_1, it didn't stay still during the loop. Halfway through the loop, v_1 was multiplied by ω^a. This branch-shift altered the inner function $f(v_1)$, which forced the outer radical v_2 to jump to a different branch. Therefore, at the end of γ_1, v_2 might be multiplied by some root of unity ζ^c.

To fix this, we nest the commutator. We build two separate 1-level commutator loops, A and B.

> As proven above, loop A leaves all Level 1 radicals completely unchanged at its conclusion. It can only multiply v_2 by ζ^c.
>
> Loop B also leaves all Level 1 radicals completely unchanged at its conclusion. It can only multiply v_2 by ζ^d.

Now, we construct the 2-level nested commutator: $\gamma_2 = [A, B] = A \cdot B \cdot A^{-1} \cdot B^{-1}$

Because Level 1 radicals (v_1) evaluate to a multiplier of 1 at the end of both A and B, they act like fixed, single-valued constants relative to these loops. Therefore, the multipliers ζ^c and ζ^d experienced by v_2 are just simple, independent complex numbers. They commute perfectly: $\zeta^c \cdot \zeta^d \cdot \zeta^{-c} \cdot \zeta^{-d} = 1$

The Result: The 2-level nested commutator has now neutralized Level 2 radicals. After running γ_2, both Level 1 and Level 2 radicals are forced back to their starting values.

4. The Inductive Collapse of the Tower This is a perfect inductive trap. Every time you add another layer of commutator nesting, you neutralize the next corresponding tier of the radical tower:

Loop Structure		Neutralized Radicals
Base Loops (P, Q)	$\longrightarrow$	None (All branches twist)
1-Level $[P, Q]$	$\longrightarrow$	Level 1 Radicals return to 1
2-Level $[[P, Q], [R, S]]$	$\longrightarrow$	Level 2 Radicals return to 1
3-Level $[[[P, Q], [R, S]], [[T, U], [V, W]]]$	$\longrightarrow$	Level 3 Radicals return to 1
$\vdots$		$\vdots$

If a quintic formula uses a tower of radicals that goes k layers deep, a k-level nested commutator loop will force every single radical in that entire formula to evaluate back to a multiplier of 1. The entire multi-tiered algebraic expression is stripped of its multi-valued nature and behaves like a rigid, single-valued rational function. Yet, because the roots of the quintic belong to the perfect group A_5, a k-level nested commutator of 3-cycles still physically permutes the actual roots. The algebra has been frozen solid by the nesting, but the geometry refuses to stop moving – proving that the formula and the roots can never match.

14 Monodromy

Explain the natural covering responsible for the complexity of the ramification of roots of the general real polynomial equation.

The complexity of how the roots of a general polynomial equation branch, merge, and change is governed by a fundamental geometric structure: the natural covering map of configuration spaces.

When looking at real polynomials, this geometry explains why formulas for roots (like the quadratic formula) become impossible for polynomials of degree 5 or higher.

Here is the step-by-step breakdown of this natural covering and how it creates ramification complexity.

1. The Space of Ordered Roots vs. Unordered Roots

To understand the covering, imagine a general polynomial of degree n with distinct roots. We define two different geometric spaces:

> The Root Space (X): The space of all possible ordered lists of n distinct roots $(x_1, x_2, \ldots, x_n)$. Because the roots must be distinct, we take the entire space $\mathbb{R}^n$ (or $\mathbb{C}^n$ for the complex case) and subtract all the hyperplanes where any two roots are equal $(x_i = x_j)$.
>
> The Coefficient Space (Y): The space of the actual polynomials themselves, represented by their coefficients. A polynomial is uniquely defined by its coefficients, provided it has no repeated roots.

2. The Natural Covering Map The link between these two spaces is the Vièta map $(p : X \to Y)$.

This map takes the ordered roots and multiplies them out to find the coefficients of the polynomial:

$$p(x_1, \ldots, x_n) = (a_{n-1}, \ldots, a_0)$$

Because changing the order of the roots (e.g., swapping x_1 and x_2) results in the exact same polynomial, the map p is a finite covering map. Specifically:

Every single point in the coefficient space Y (a polynomial with distinct roots) sits directly underneath exactly $n!$ (n-factorial) points in the root space X.

The symmetric group S_n acts on the roots by permutation, shuffling the sheets of this covering space without changing the polynomial below.

3. The Source of Ramification: The Discriminant The covering map is perfectly neat and smooth only when the roots are completely distinct. Complexity explodes when we approach polynomials with repeated roots.

The Discriminant Variety (Δ): This is the boundary where the polynomial has at least one multiple root (where the discriminant equals zero).

As coefficients move through space Y and cross or wrap around the discriminant Δ, the roots are forced to collide and branch.

In the real numbers, crossing the discriminant changes the number of real roots (e.g., two real roots collide, vanish, and become two complex conjugate roots).

4. Monodromy and the Braid Group The topological complexity of this covering is tracked by moving coefficients in a continuous loop around the discriminant variety without touching it.

Start with a polynomial with distinct roots.

Continuously vary its coefficients in a loop that winds around the discriminant.

When the loop returns to the exact starting polynomial, the roots will have returned to their original positions – but permuted.

This phenomenon is called monodromy. In the complex plane, tracing these loops forms the Artin Braid Group. The braids show exactly how the sheets of the covering space twist around each other.

Summary of Complexity The "natural covering" is a sheeted landscape where the top layer represents known, ordered roots, and the bottom layer represents the polynomial's coefficients. The complexity of ramification arises because loops in the coefficient space lift to paths that jump between different sheets, permuting the roots. Vladimir Arnold used the cohomology of these twisted spaces to prove that certain algebraic functions are topologically too complex to ever be simplified by standard algebraic transformations.

1. The Core Premise: Radicals Are Locally Predictable Loops To understand why radicals fail, we must first look at what a radical (k-th root) does topologically.
Imagine the function $w = \sqrt[k]{z}$. If you start at $z = 1$ and move z in a continuous loop around the origin exactly once, w does not return to its starting value. Instead, it lands on a different "sheet" of the function (multiplied by a root of unity, $e^{2\pi i/k}$).

The Rule of Radicals: Tracing a loop around a branch point k times will always bring a k-th root radical back to its starting sheet.

Because of this, if you compute a loop called a commutator ($A \cdot B \cdot A^{-1} \cdot B^{-1}$), any radical function inside that loop tracks outward and inward along the sheets, perfectly canceling itself out and returning to its original value.

2. The Monodromy Group: The Permutation of Roots As established by the natural covering space, if we take a general polynomial of degree n and move its coefficients in a loop around the discriminant variety, the roots track along paths and swap places.

The set of all possible ways roots can swap places by tracing loops is called the Monodromy Group. For a general polynomial of degree n, the monodromy group is the full symmetric group S_n (all possible permutations of n elements).

3. The Algebraic vs. Topological Matchup

An algebraic formula using only addition, subtraction, multiplication, division, and radicals corresponds to a very specific topological property: the monodromy group must be solvable.
Topologically, a solvable group means that if you keep taking loops of loops (nested commutators), the twisting eventually untangles and collapses to nothing.

For degrees $n \leq 4$: If you take a loop of loops, the permutations of the roots eventually cancel out. The twistiness of the space is simple enough that nesting radicals can perfectly mirror and "undo" the swapping of the roots.

For degrees $n \geq 5$: The symmetric group S_n contains the alternating group A_n, which is a simple group (specifically, it cannot be broken down further, and its commutator subgroup is itself).

4. The "Unsplittable Knot" of the Quintic Here is the geometric death blow to the quintic formula for $n = 5$:

Because A_5 is equal to its own commutator subgroup, you can create a complex loop of coefficient movements (a commutator loop) that completely untangles any nested radical formula you could possibly write. Tracing this loop ensures every radical in your proposed formula returns to its original sheet.

However, because A_5 is non-abelian and simple, tracing that exact same loop of coefficients does not return the roots of the polynomial to their original positions. The roots are still permuted (e.g., root 1 is now where root 2 was).

This creates an impossible contradiction. If a quintic formula existed, traveling along this specific loop would mean the formula outputs its original value, but the actual roots of the polynomial have changed.

Conclusion A formula using radicals can only navigate covering spaces whose sheets twist in a hierarchical, solvable way. For polynomials of degree 5 or higher, the natural covering space wraps around the discriminant variety in a complex, interlocking web (S_5). Topologically, the roots are knotted together so tightly that the simple unlooping mechanism of radical branches is fundamentally incapable of separating them.

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